Software Engineering and Digital Arts Conservation (SEDA)

Art History

Inge Hinterwaldner (unibz)

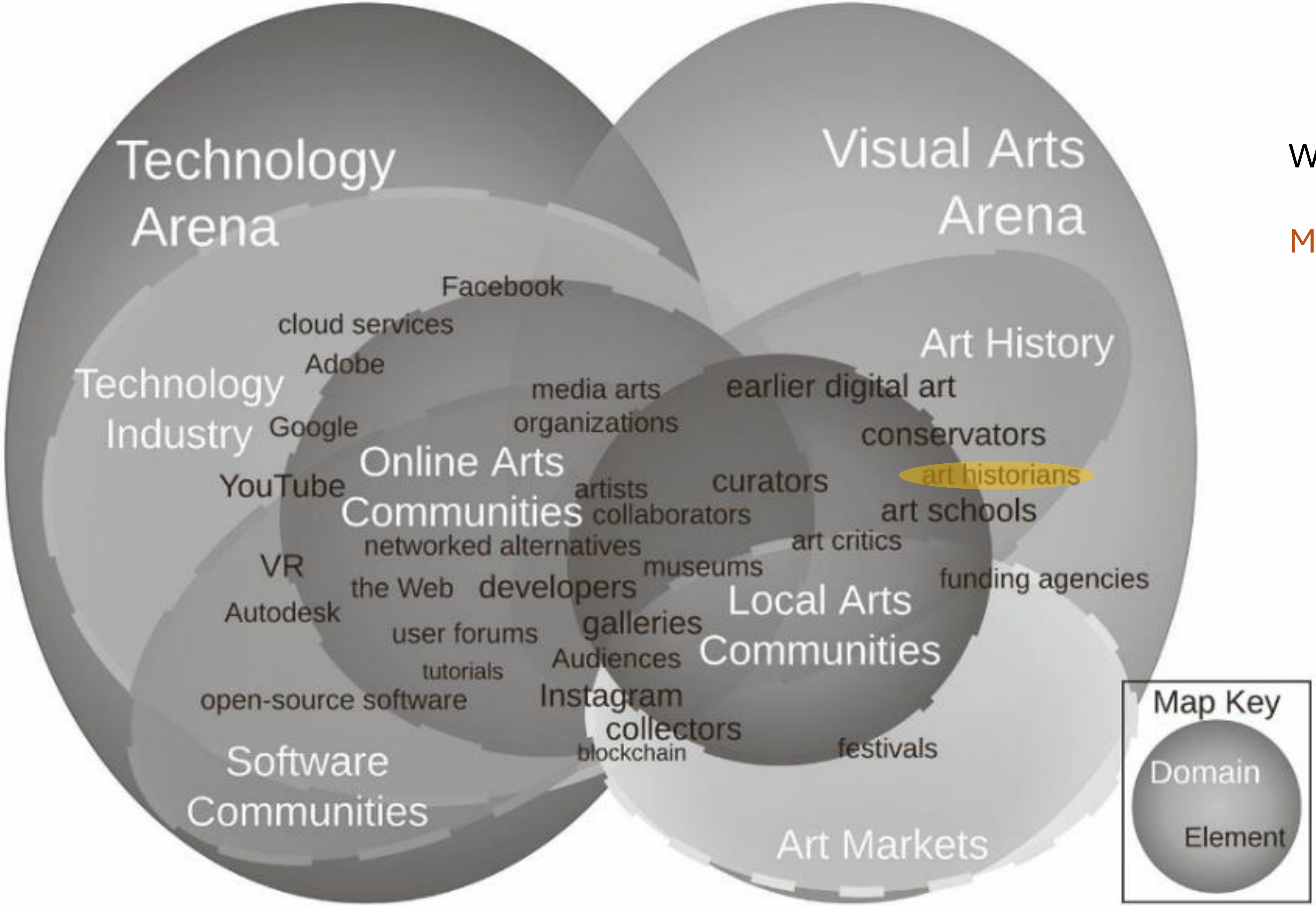
Workshop, U Amsterdam, 24.3.2026

DFG Deutsche Forschungsgemeinschaft

European Research Council
Established by the European Commission

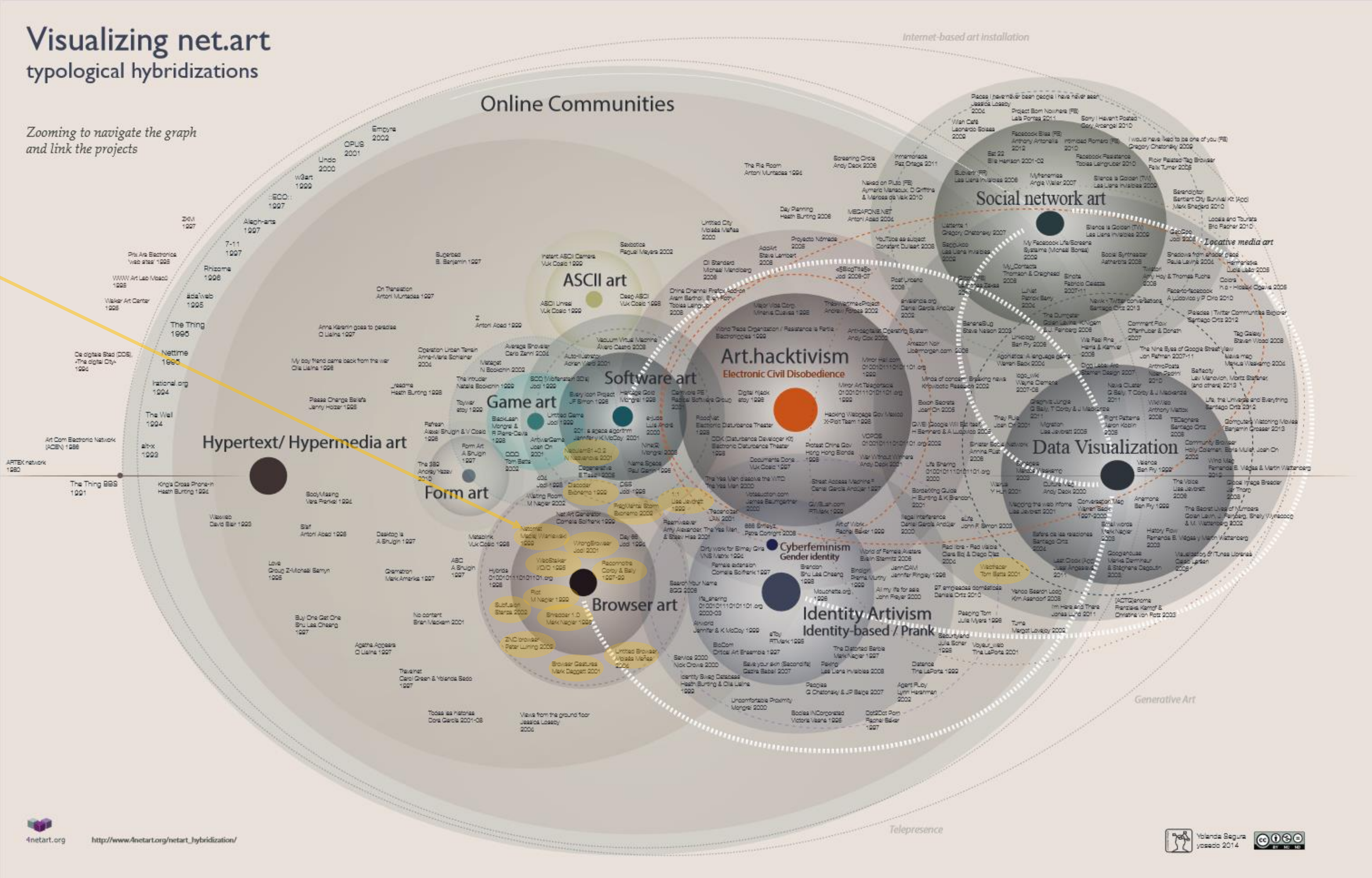
What are art historians good for?

Model building (maybe?)

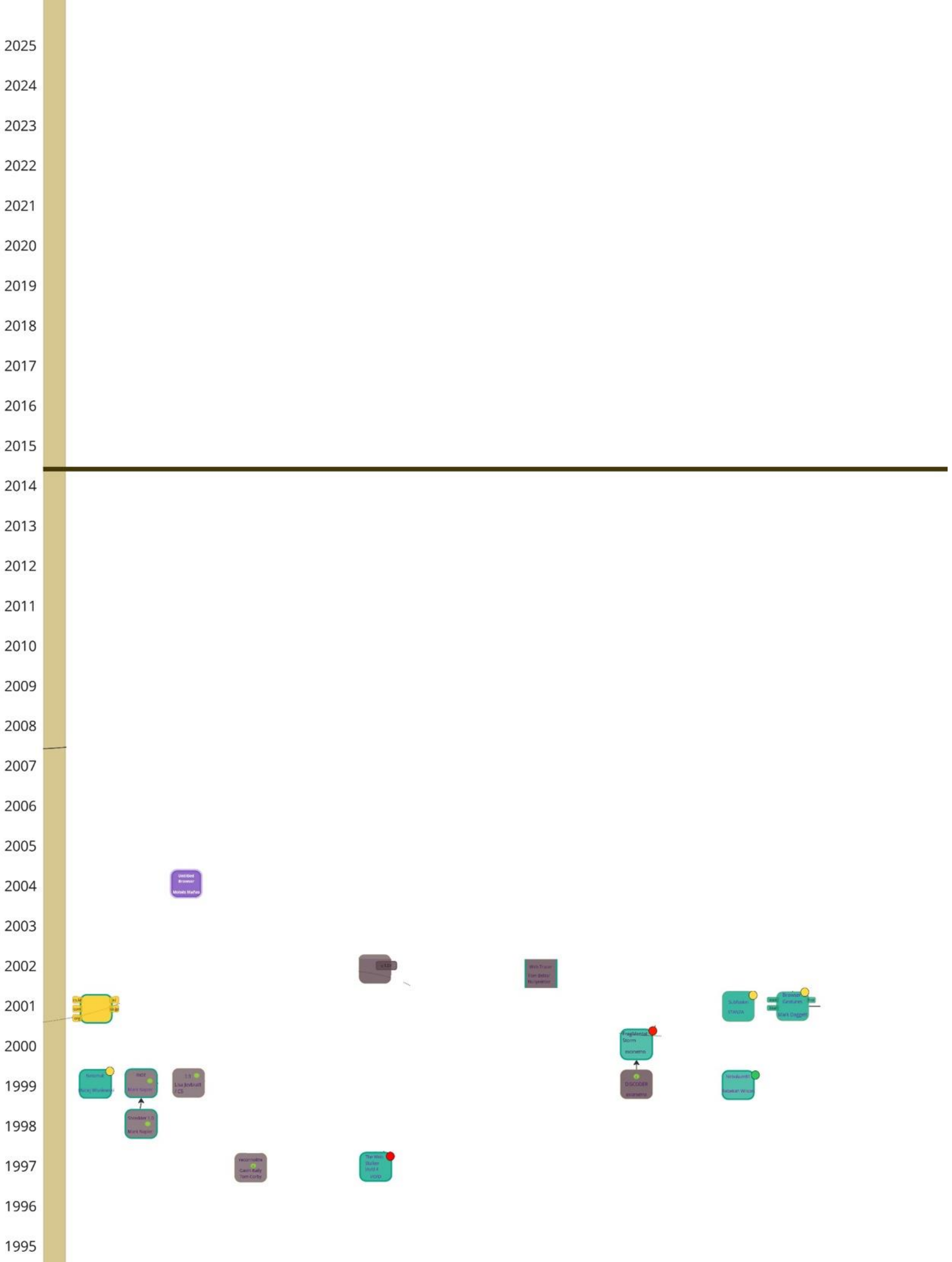


Social worlds and arenas map of artists and curators involved in the care of net-based art.
Source:
Colin Post: The Art of Digital Curation: Co-operative Stewardship of Net-Based Art. Archivaria, (92), 2001, pp. 6–47.
https://doi.org/10.7202/1084738ar

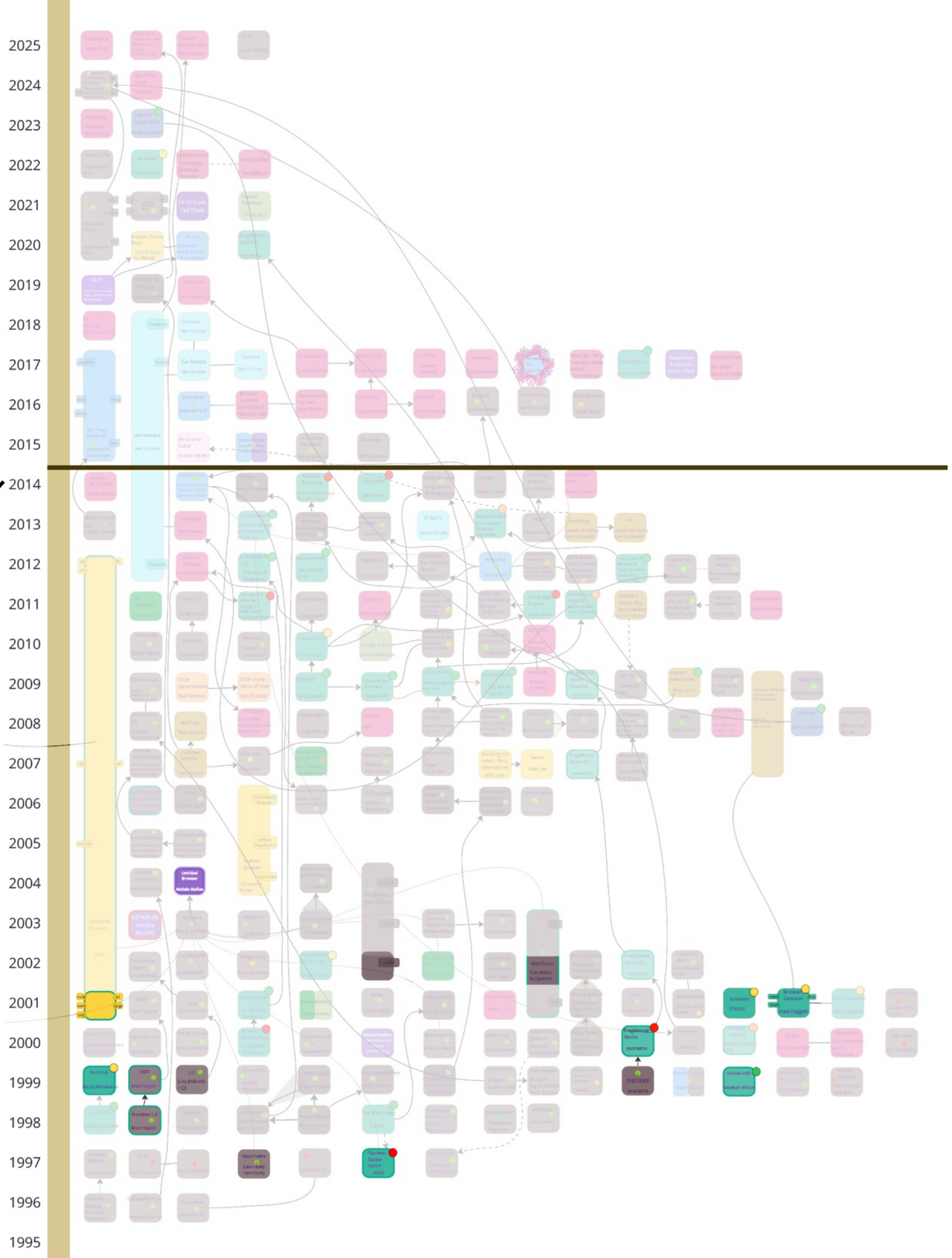
… here is the state of the art 2014 in another survey diagram of net art: known artistic browsers: 15



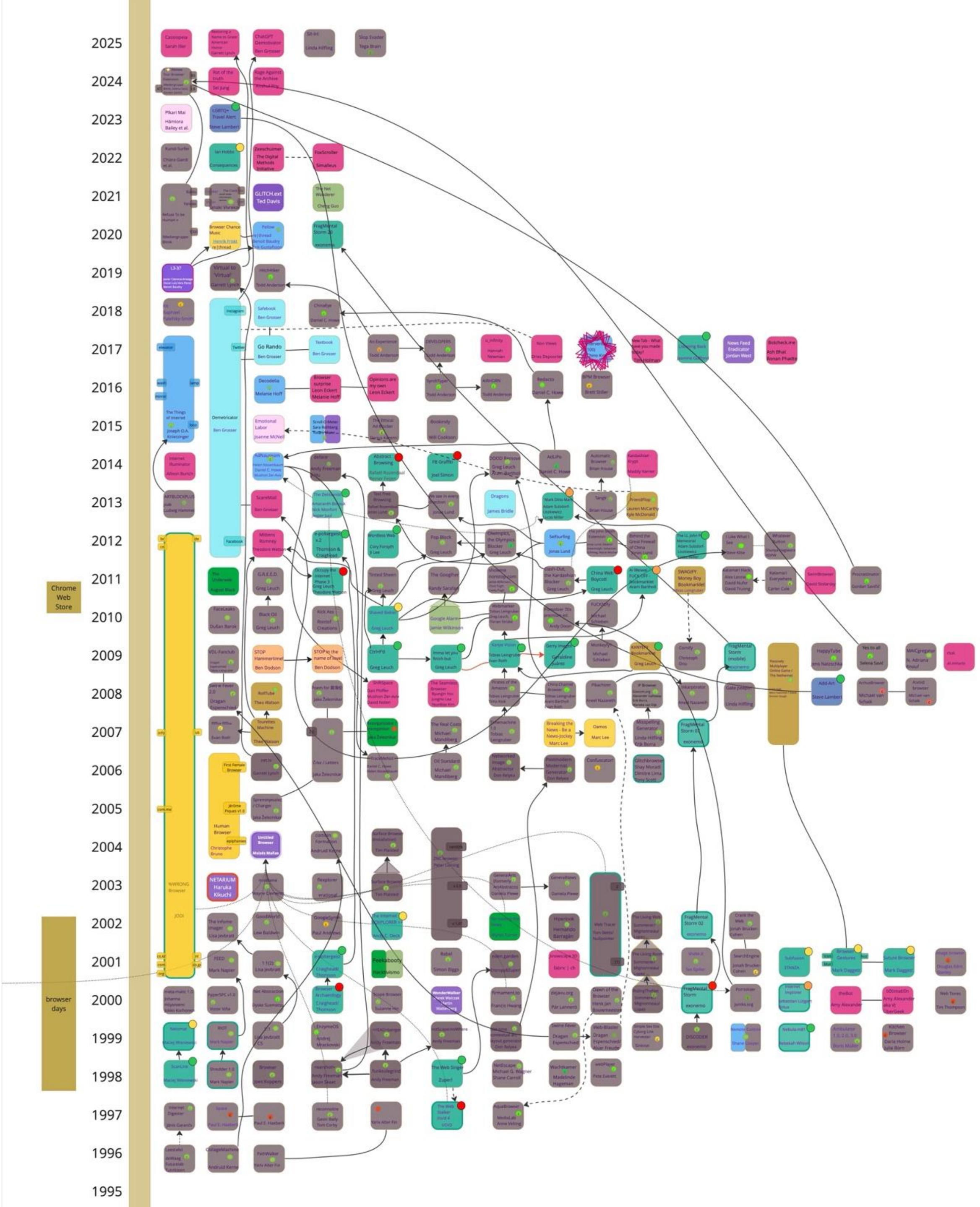
Yolanda Segura: Visualizing net.art, 2014. Source: http://www.4netart.com/org/netart_hybridization/hibridaciones_netart_yosedo.pdf



Yolanda Segura: Visualizing net.art, 2014. Source: http://www.4netart.com/org/netart_hybridization/hibridaciones_netart_yosedo.pdf



Yolanda Segura: Visualizing net.art, 2014. Source: http://www.4netart.com/org/netart_hybridization/hibridaciones_netart_yosedo.pdf


2025
2024
2023
2022
2021
2020
2019
2018
2017
2016
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2012
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2008
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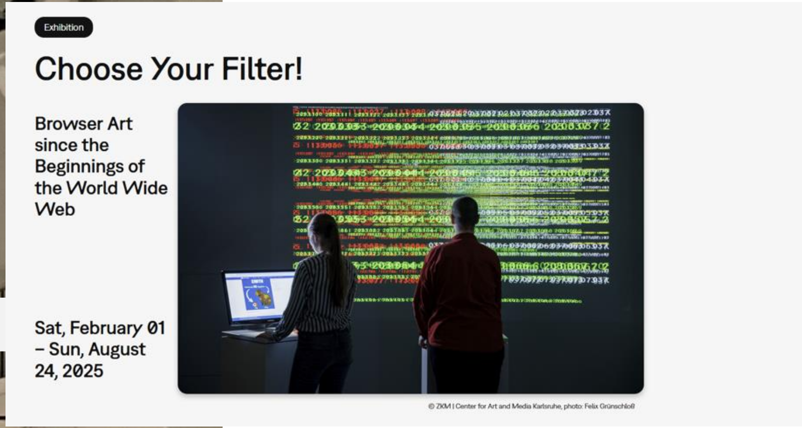
2006
2005
2004
2003
2002
2001
2000
1999
1998
1997
1996
1995
Chrome Web Store
browser days

March 2026


Choose Your Filter!
Browser Art seit den Anfängen
des World Wide Web
Noch bis
24.8.25

Show works as often as possible



software art

paintings

Show works as little as possible

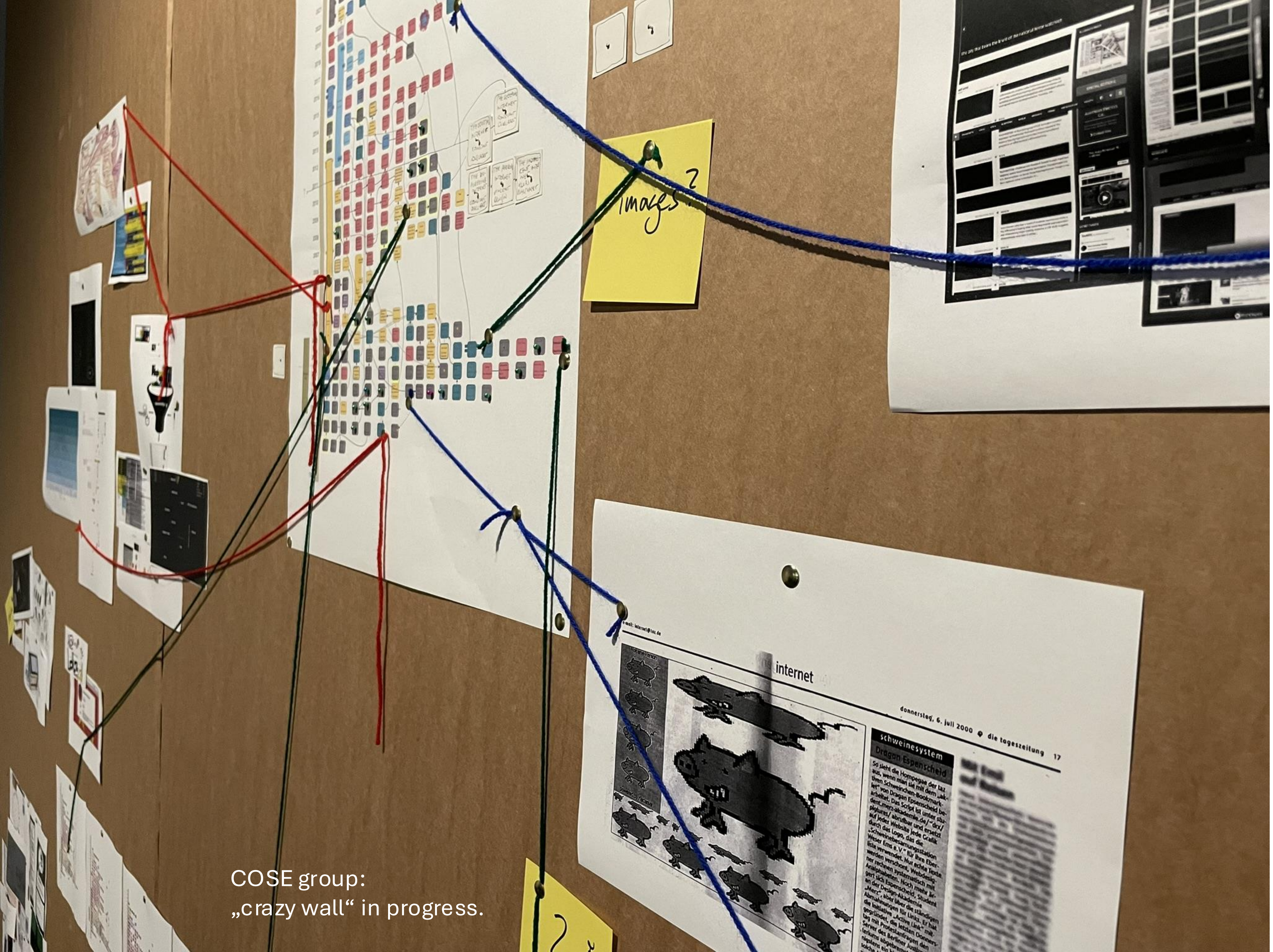
2 simultaneous exhibitions at ZKM 2025 –
2 very different conservation strategies

Choose Your Filter! Browser Art since the Beginnings of the World Wide Web, ZKM, Feb 1-Aug 28, 2025

survey of a field

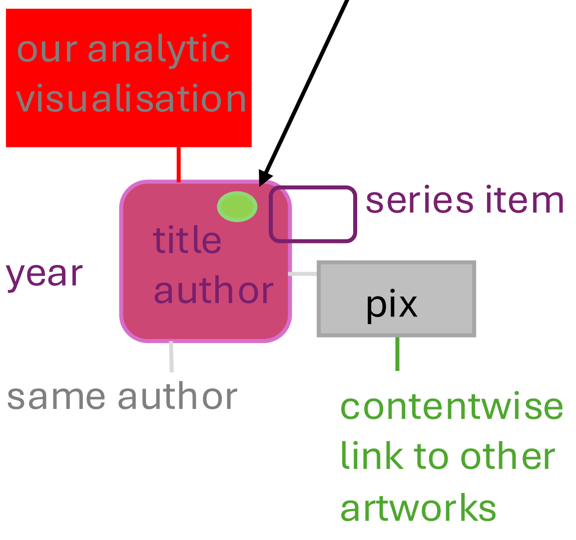
Timeline / Database (audiovisual, poster, book)
+ source/reference
(COSE: *Browser Art Browser*, 2025: based on Jeffrey Shaw: *Linear Browser*, 1999)

35:3



COSE group:
„crazy wall“ in progress.

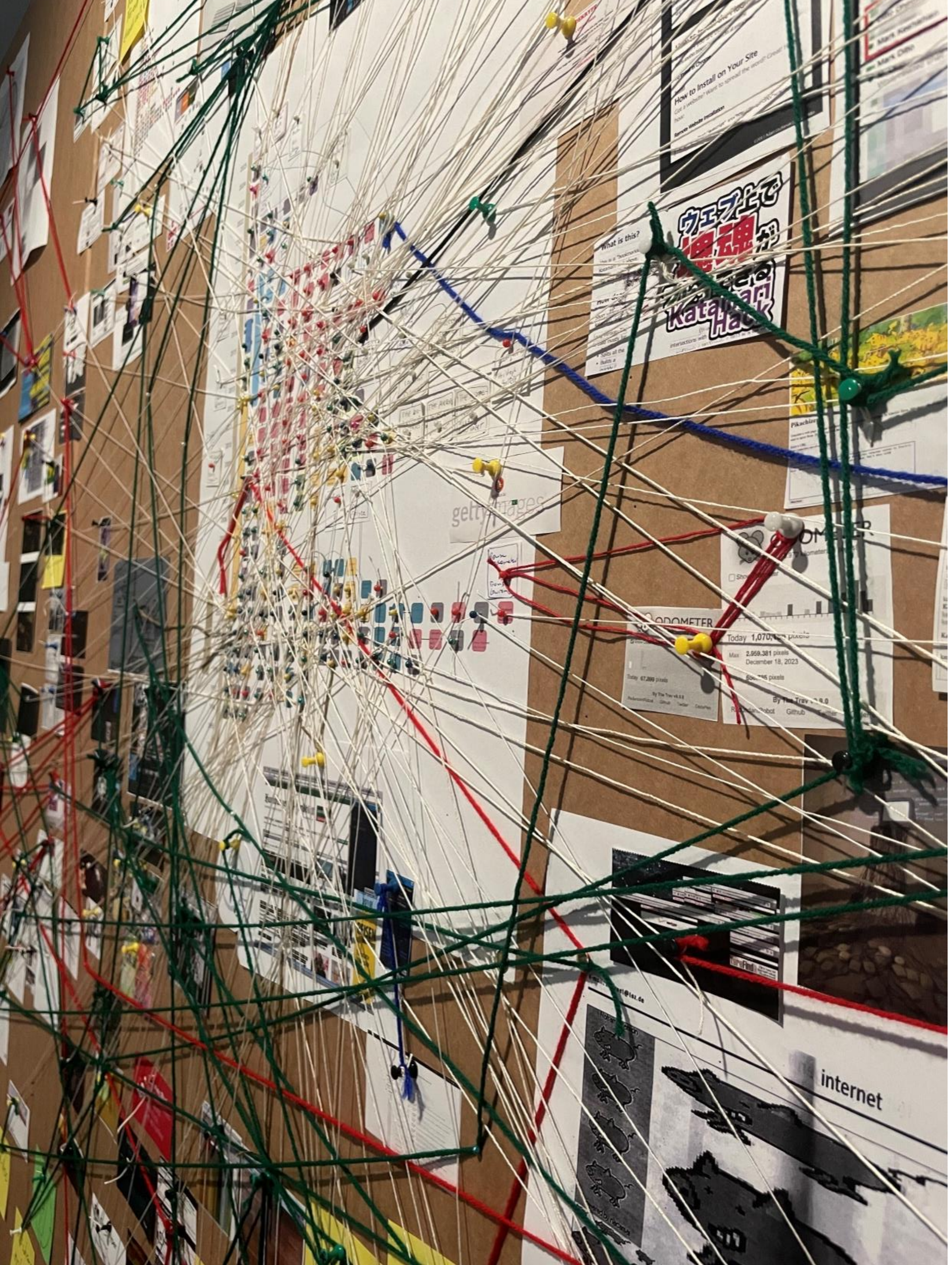
colour code + circle:
status of research (interview)
and of publication
(catalogue entry)



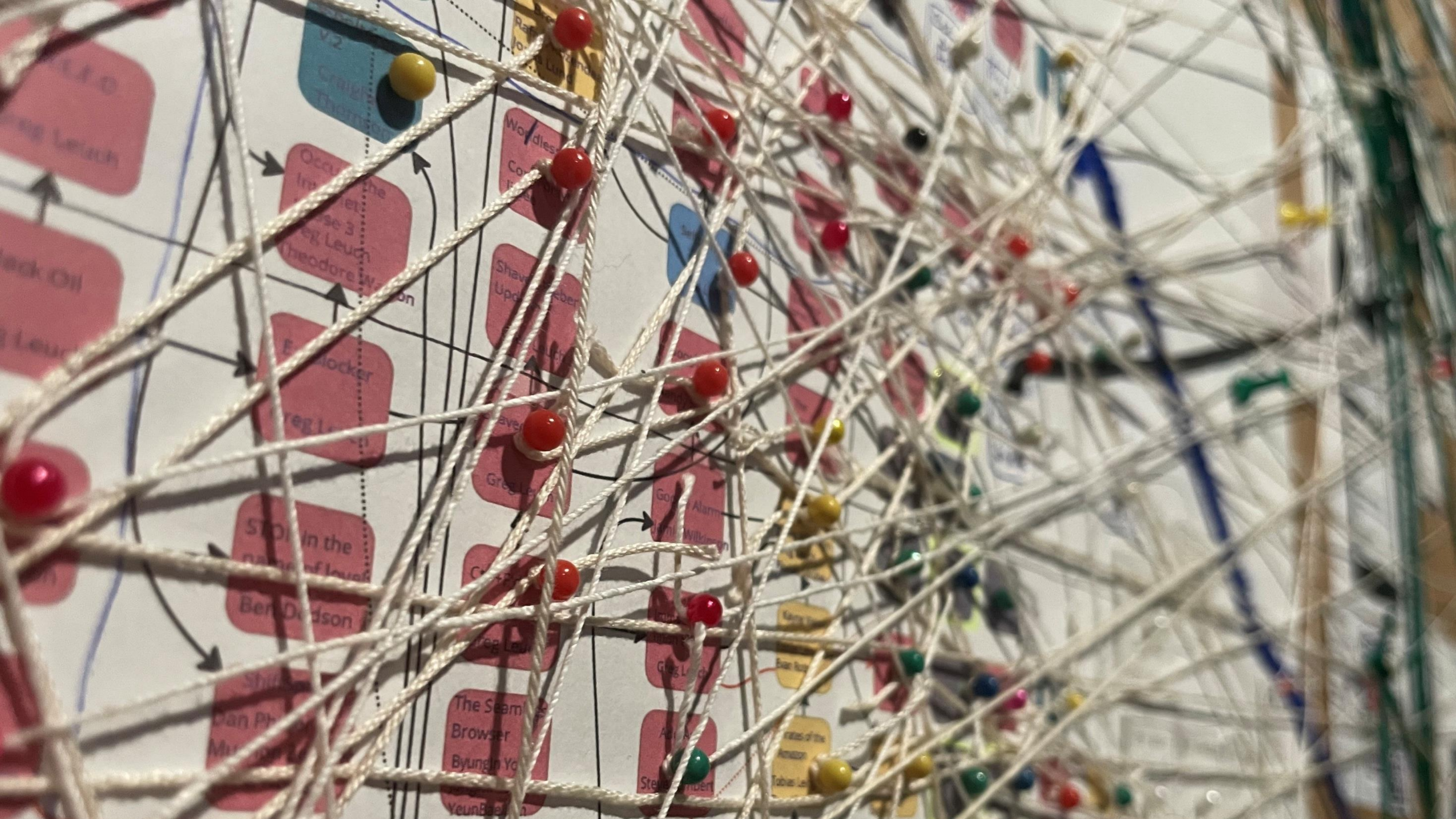
documentation of:
past activities

current activities

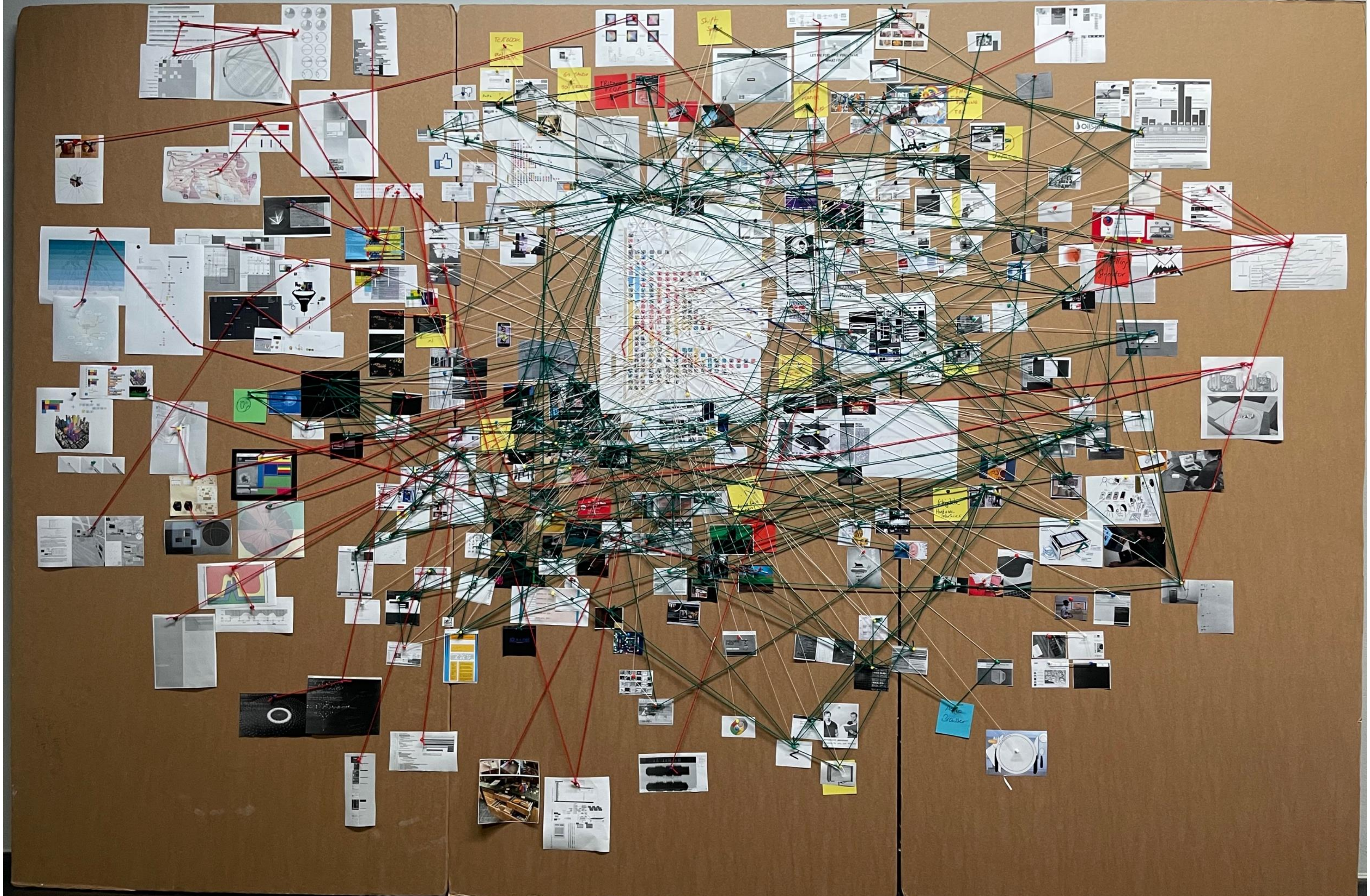
emergent feats:
future activities


How to Install on Your Site
What is this?
gettyimages
ODOMETER
internet

NEWS
karp-karp-karp-karp


V.2
Browser


ThinkPad

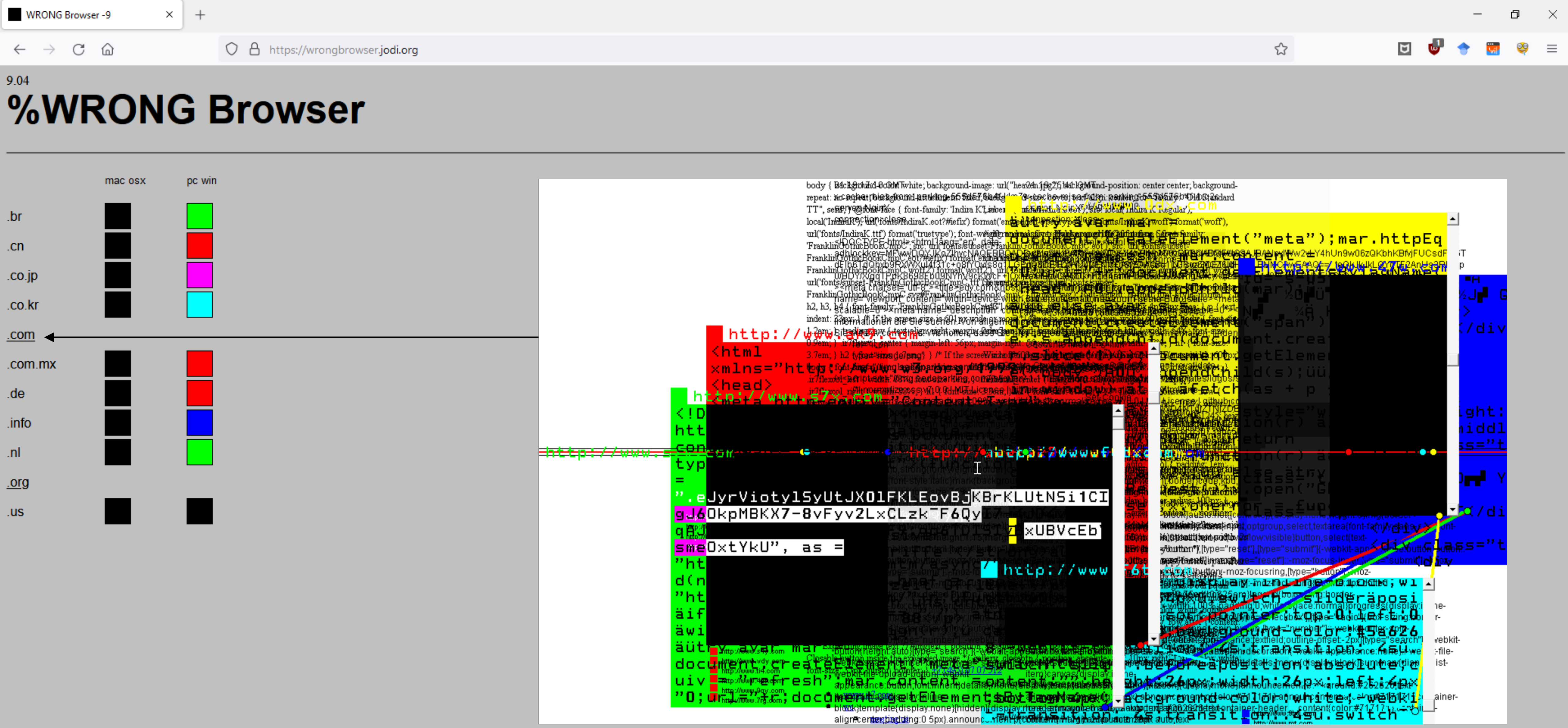


survey of a field

COSE: Crazy wall,
State of the art:
22.1.2026.

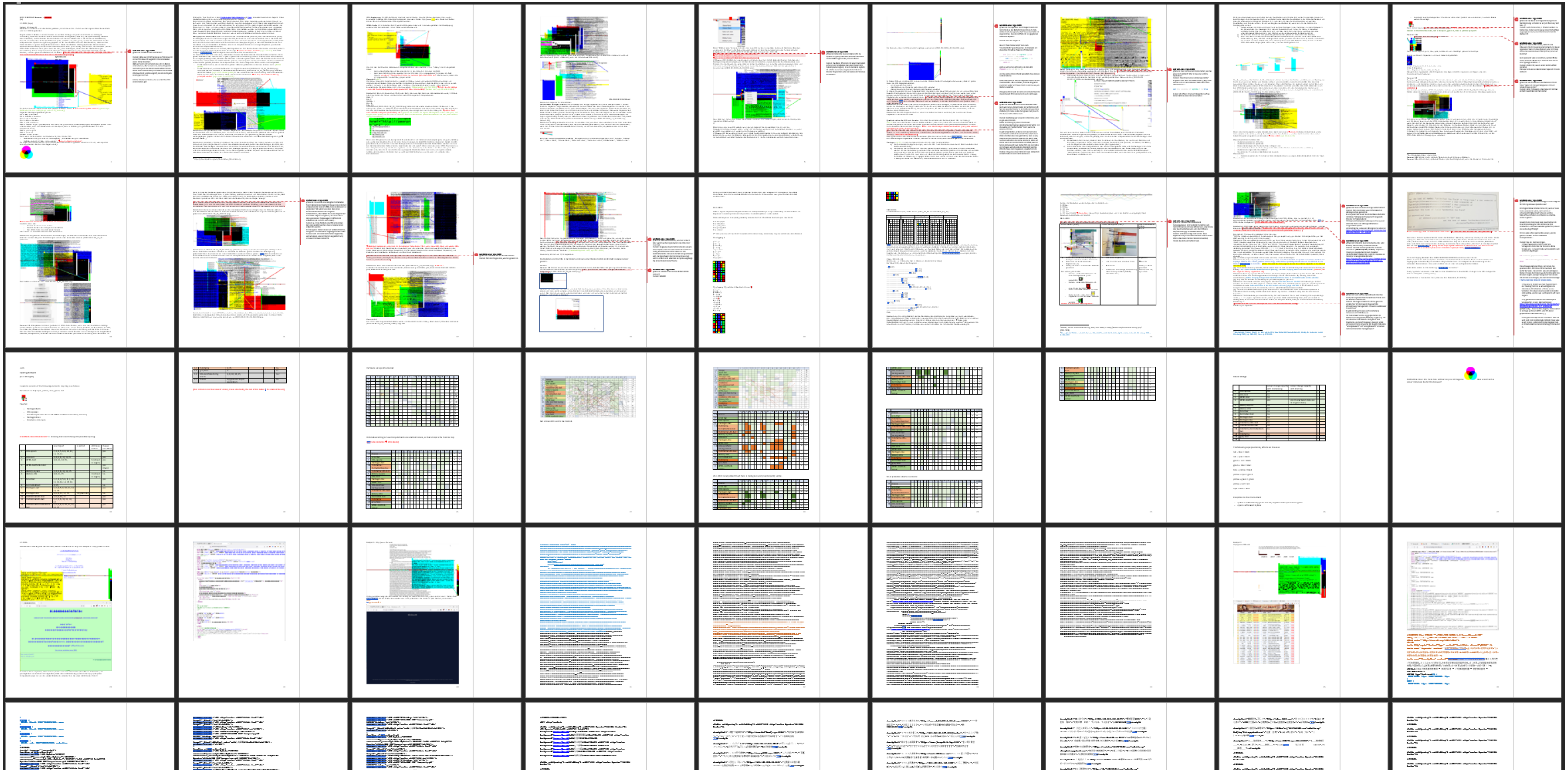
Visualising
semantic fields,
the cross-
referencing and
the catalogue
-
living document

With about 220 artistic web browsers we considerably expanded what was known of that genre ...

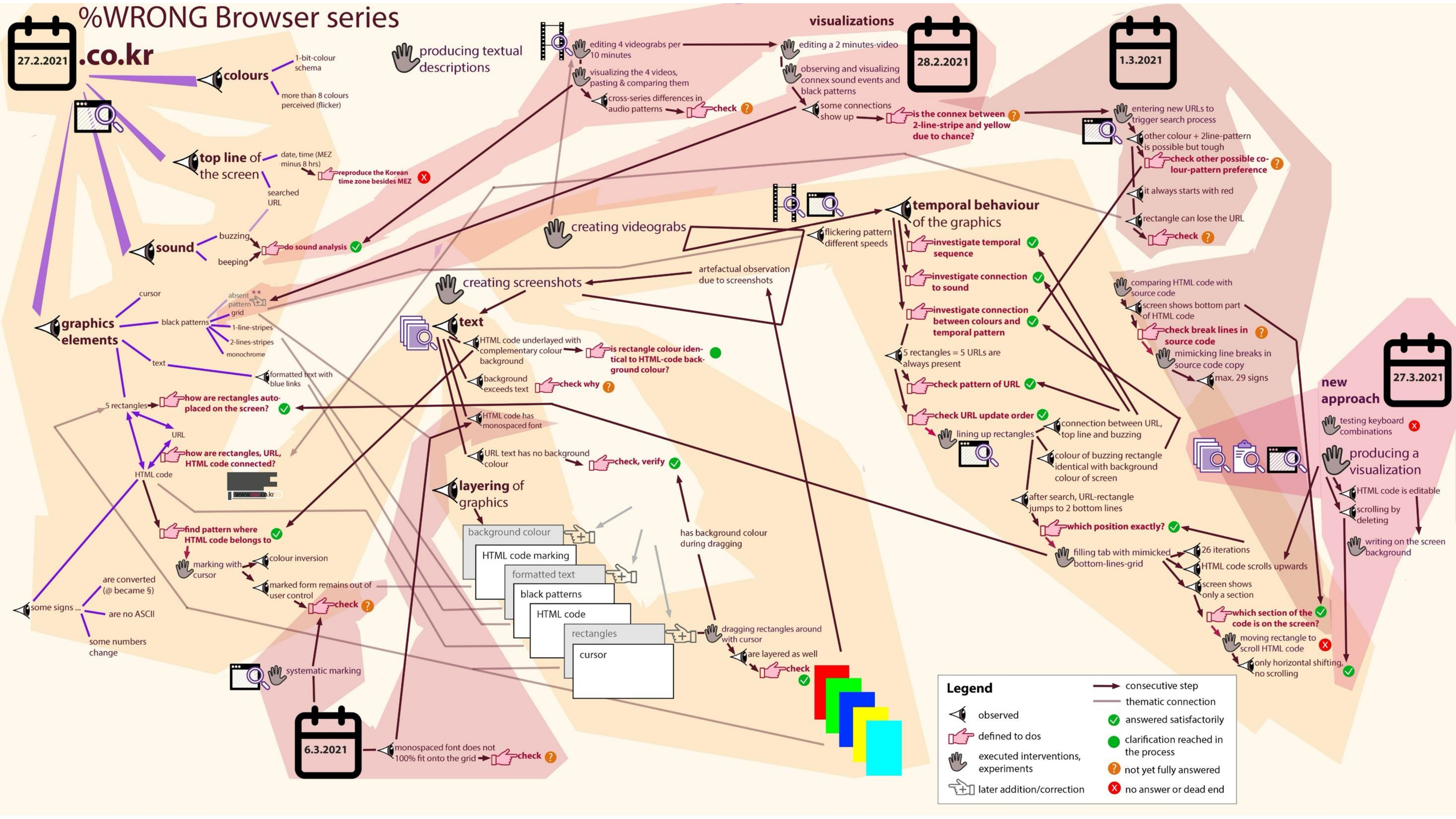


JODI: %WRONG Browser .com. screenshot

JODI: %WRONG Browser series,
2001-2012. Download menu.

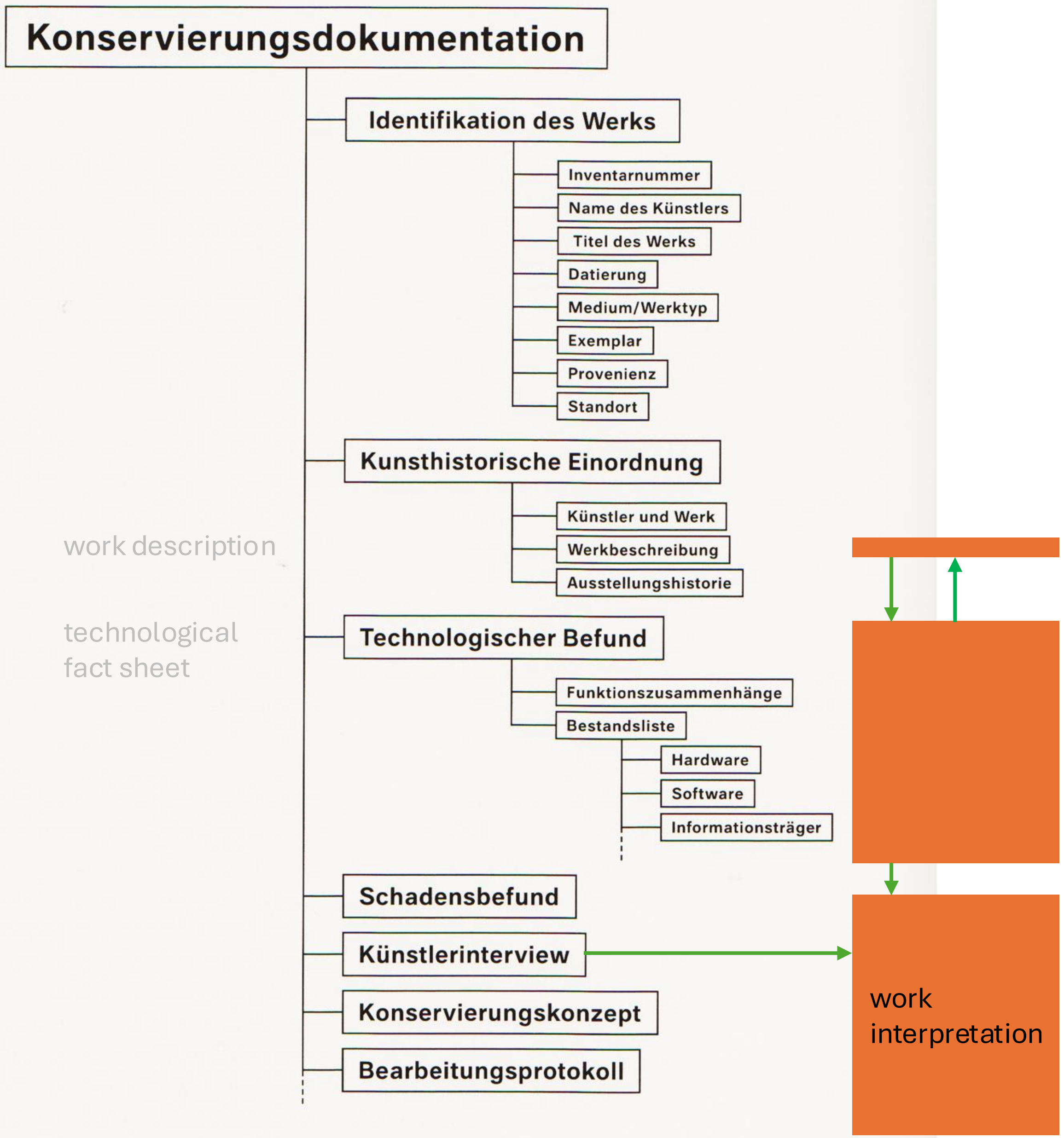


Semi-structured documentation of *.com* (mix of text, tabs, screenshots, diagrams, comparisons, comments …) in a word document

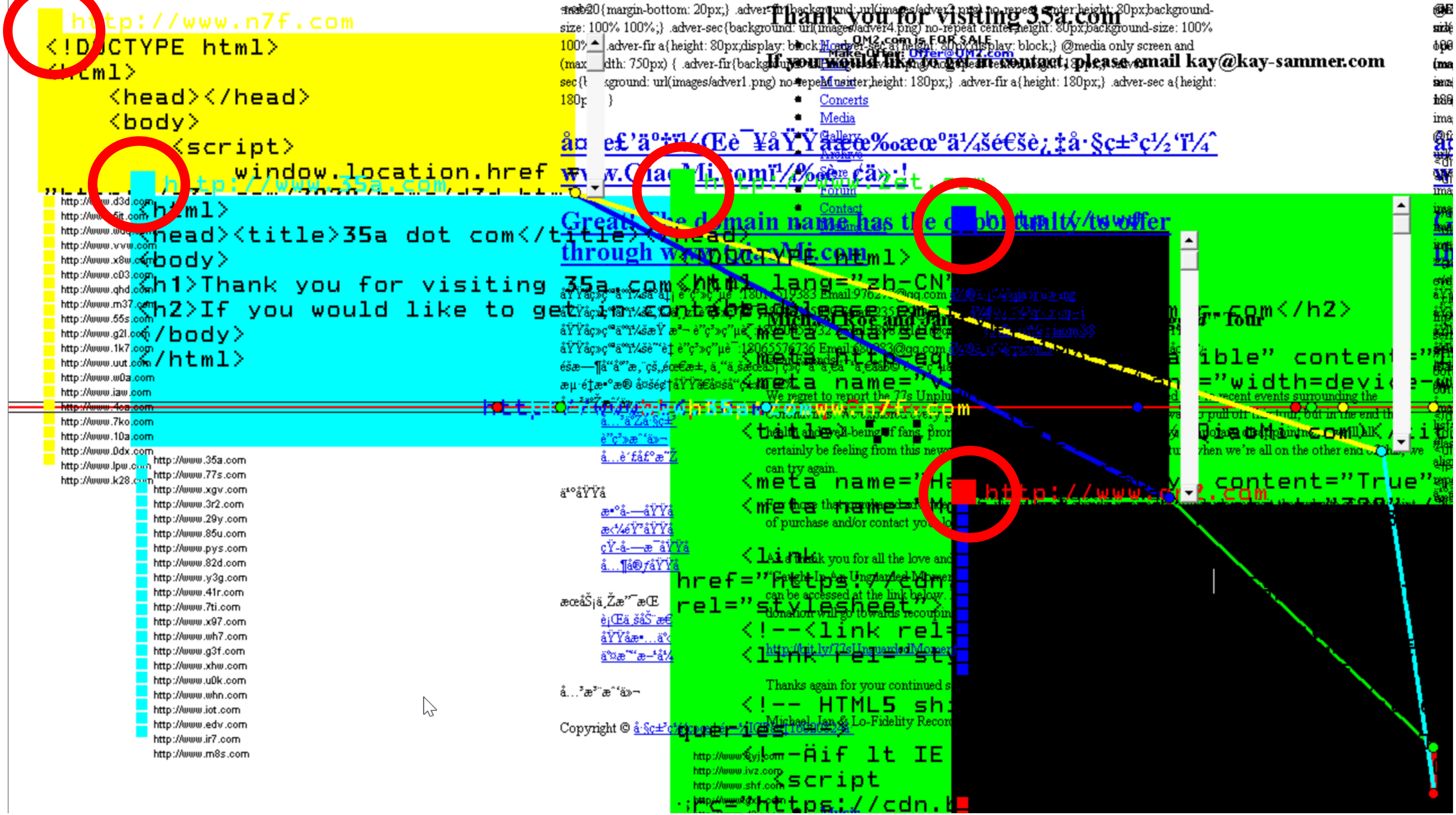


%WRONG Browser series
.co.kr
27.2.2021
colours
1-bit-colour schema
more than 8 colours perceived (flicker)
top line of the screen
date, time (MEZ minus 8 hrs)
reproduce the Korean time zone besides MEZ
searched URL
sound
buzzing
beeping
do sound analysis
graphics elements
cursor
black patterns
absent pattern
grid
1-line-stripes
2-lines-stripes
monochrome
text
formatted text with blue links
5 rectangles
how are rectangles auto-placed on the screen?
URL
how are rectangles, URL, HTML code connected?
HTML code
www.[redacted]
find pattern where HTML code belongs to
marking with cursor
colour inversion
marked form remains out of user control
check
some signs ...
are converted (@ became §)
are no ASCII
some numbers change
systematic marking
6.3.2021
monospaced font does not 100% fit onto the grid
check
producing textual descriptions
editing 4 videograbs per 10 minutes
visualizing the 4 videos, pasting & comparing them
cross-series differences in audio patterns
check
visualizations
editing a 2 minutes-video
observing and visualizing connex sound events and black patterns
some connections show up
is the connex between 2-line-stripe and yellow due to chance?
28.2.2021
1.3.2021
entering new URLs to trigger search process
other colour + 2line-pattern is possible but tough
check other possible colour-pattern preference
it always starts with red
rectangle can lose the URL
check
creating videograbs
creating screenshots
artefactual observation due to screenshots
text
HTML code underlayed with complementary colour background
is rectangle colour identical to HTML-code background colour?
background exceeds text
check why
HTML code has monospaced font
URL text has no background colour
check, verify
layering of graphics
background colour
HTML code marking
formatted text
black patterns
HTML code
rectangles
cursor
has background colour during dragging
dragging rectangles around with cursor
are layered as well
check
temporal behaviour of the graphics
flickering pattern different speeds
investigate temporal sequence
investigate connection to sound
investigate connection between colours and temporal pattern
5 rectangles = 5 URLs are always present
check pattern of URL
check URL update order
lining up rectangles
connection between URL, top line and buzzing
colour of buzzing rectangle identical with background colour of screen
after search, URL-rectangle jumps to 2 bottom lines
which position exactly?
filling tab with mimicked bottom-lines-grid
26 iterations
HTML code scrolls upwards
screen shows only a section
which section of the code is on the screen?
moving rectangle to scroll HTML code
only horizontal shifting, no scrolling
comparing HTML code with source code
screen shows bottom part of HTML code
check break lines in source code
mimicking line breaks in source code copy
max. 29 signs
new approach
27.3.2021
testing keyboard combinations
producing a visualization
HTML code is editable
scrolling by deleting
writing on the screen background
Legend
observed
defined to dos
executed interventions, experiments
later addition/correction
consecutive step
thematic connection
answered satisfactorily
clarification reached in the process
not yet fully answered
no answer or dead end

Serexhe, Bernhard (ed.): digital art conservation. Konservierung digitaler Kunst: Theorie und Praxis, Ambra | V / ZKM, Vienna 2013, p. 340.

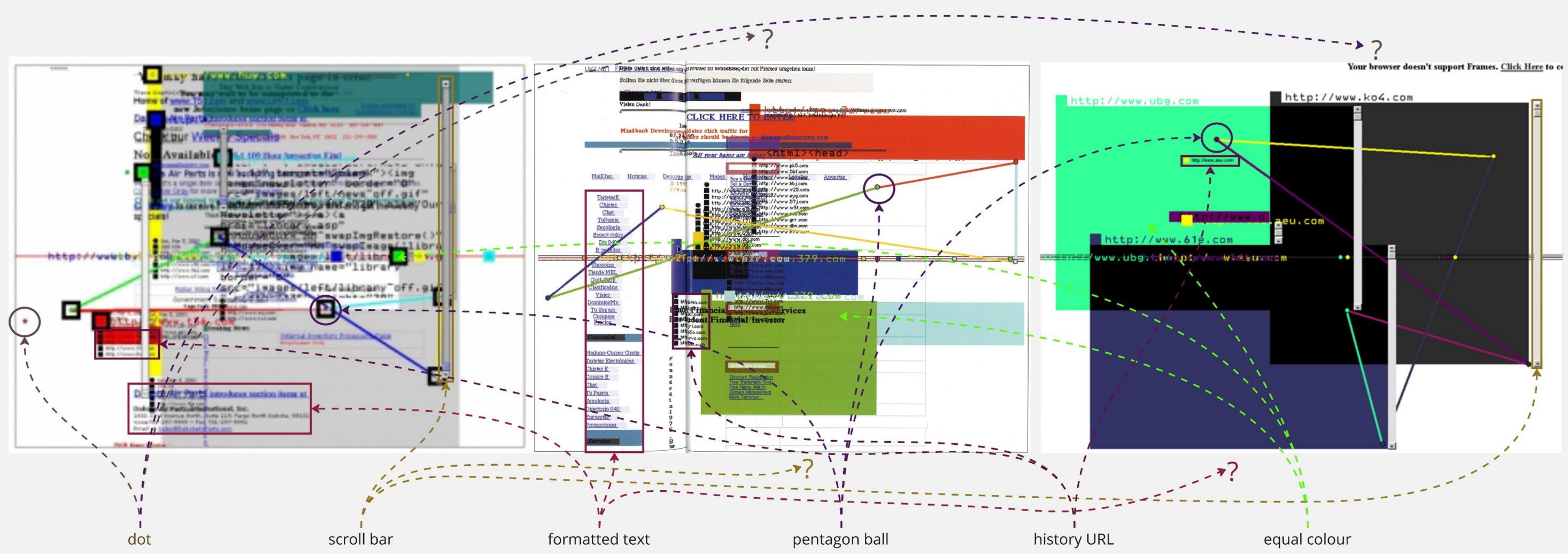


Dokumentationsstruktur
zur Aufnahme der für das
Projekt *digital art conservation*
ausgewählten Werke
Foto © ZKM | Karlsruhe



JODI: *.com*, 2001. From the series *%WRONG Browser*. Still, reworked.

The screenshot dilemma: too quick&easy to be good (enough)

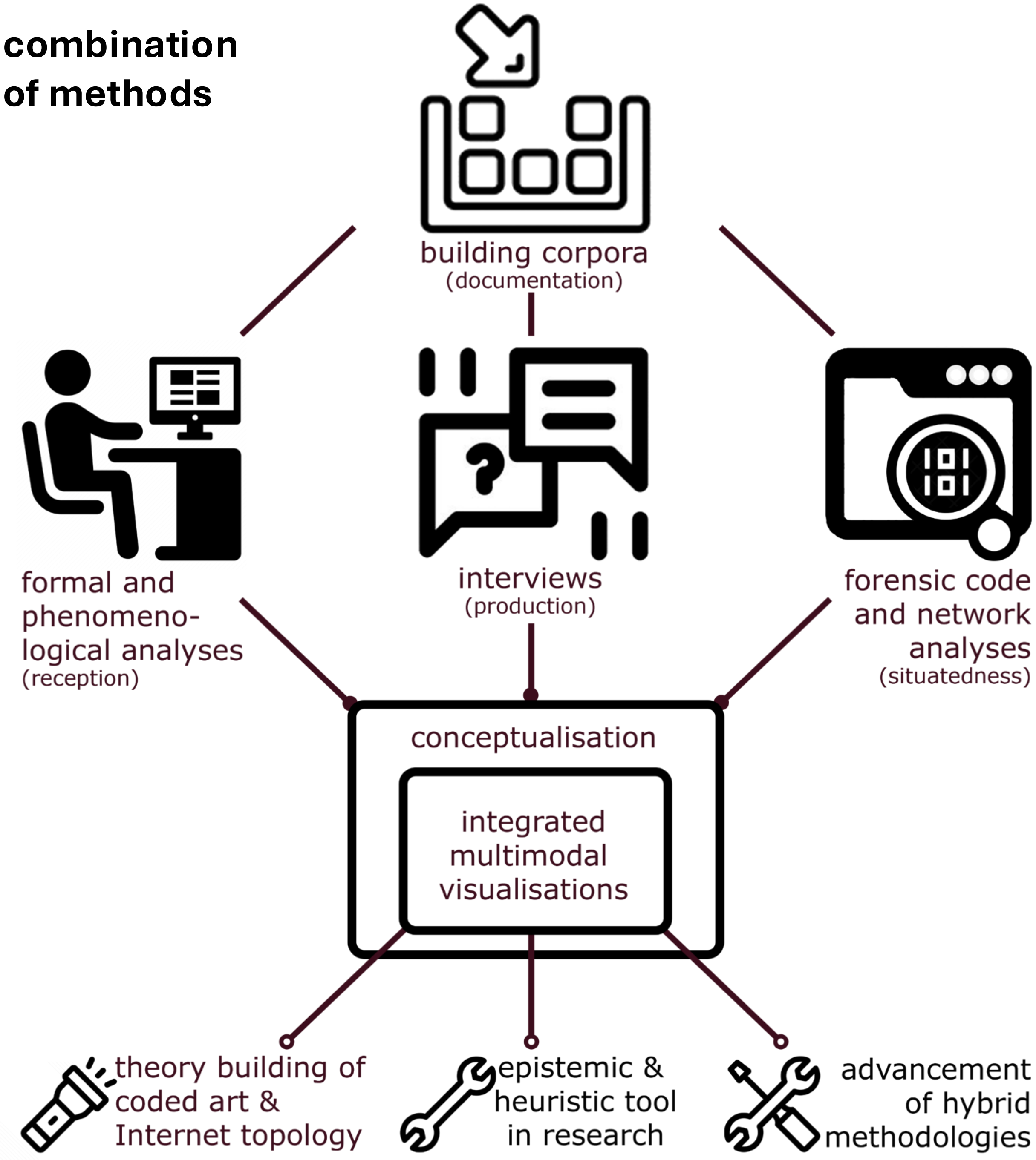


Dekker, Annet: World Wide Wrong, JODI, 24.8.2005, in: http://aaaan.net/world-wide-wrong-jodi/ (28.1.2023).

Baumgärtel, Tilman: net.art 2.0, New Materials Towards Net Art, Verlag für moderne Kunst: Nürnberg 2001, pp. 176-177.

Mackern, Brian: // navegadores web alternativos // v 2.004, in: https://netart.org.uy/browser-art/ (18.5.2023). The screenshot was taken on November 17, 2001.

combination of methods



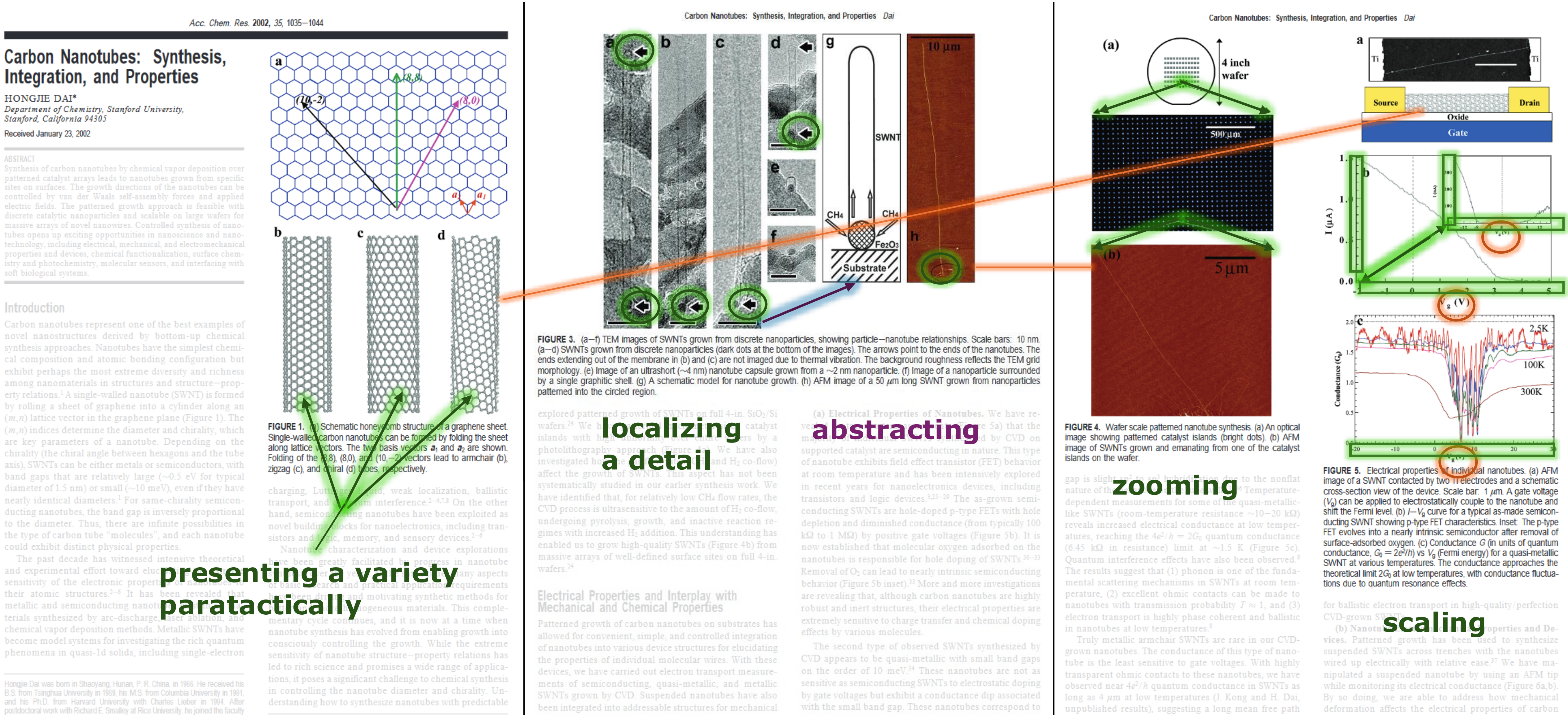
Methodology

To provide the first in-depth interpretation of this iconic browser art piece we needed to secure our knowledge base by documenting it thoroughly. We combined:

- **Interviews** (social sciences approaches)
- **Phenomenological analysis** (art history)
- **Forensic code analysis** (digital humanities, computer science, conservation)

We found the use of screenshots as being questionable and developed:

- **Model building** (arts, media design, art history)



Dai, Hongjie: Carbon Nanotubes: Synthesis, Integration, and Properties, in: Acc. Chem. Res., vol. 35, 2002, pp. 1035-1044.

Credo: rethinking of science communication is needed. Art history needs to create visualisations/models – to be more precise, to communicate better (like STEM: how to tell micro-stories) and in multiple ways. These models can evolve into a variety of forms, depending on q&a, and this broadens the accessability for different interests and publics.

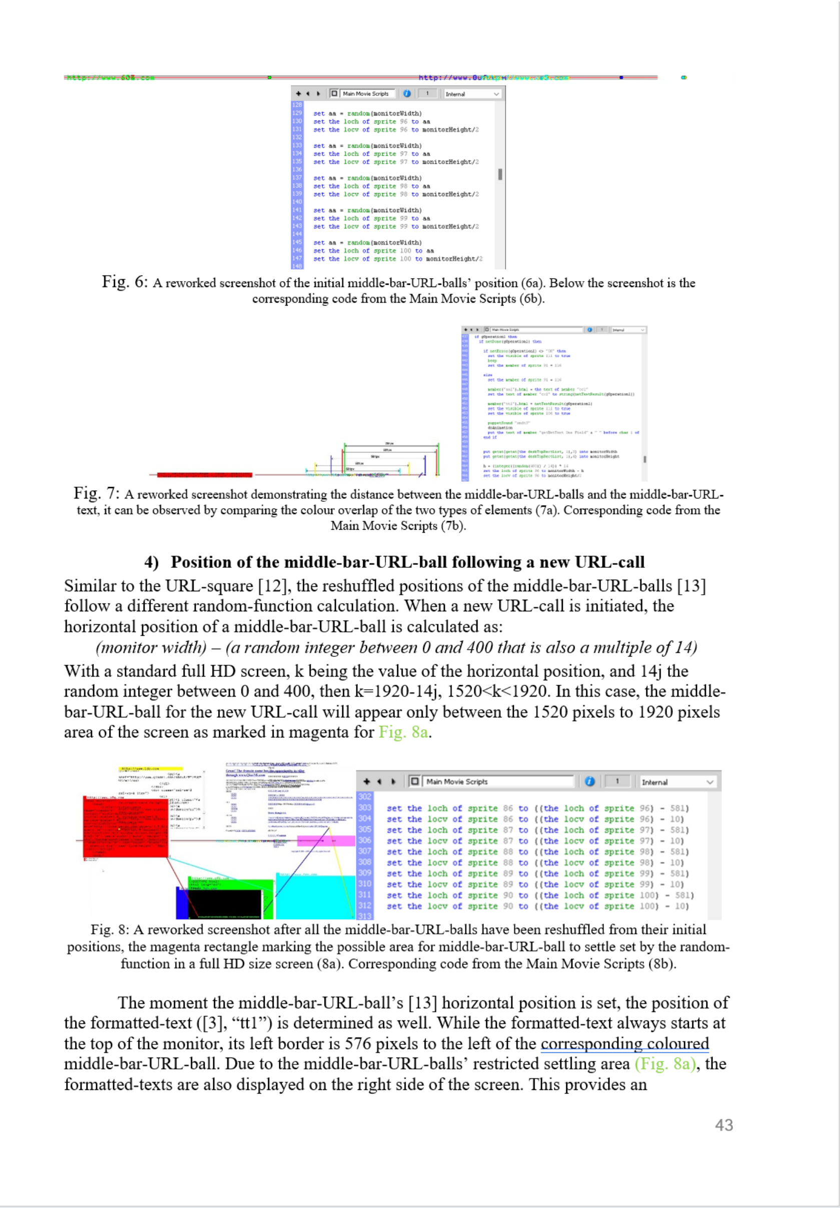


Fig. 6: A reworked screenshot of the initial middle-bar-URL-balls' position (6a). Below the screenshot is the corresponding code from the Main Movie Scripts (6b).

Fig. 7: A reworked screenshot demonstrating the distance between the middle-bar-URL-balls and the middle-bar-URL-text, it can be observed by comparing the colour overlap of the two types of elements (7a). Corresponding code from the Main Movie Scripts (7b).

4) Position of the middle-bar-URL-ball following a new URL-call

Similar to the URL-square [12], the reshuffled positions of the middle-bar-URL-balls [13] follow a different random-function calculation. When a new URL-call is initiated, the horizontal position of a middle-bar-URL-ball is calculated as:

(monitor width) – (a random integer between 0 and 400 that is also a multiple of 14)

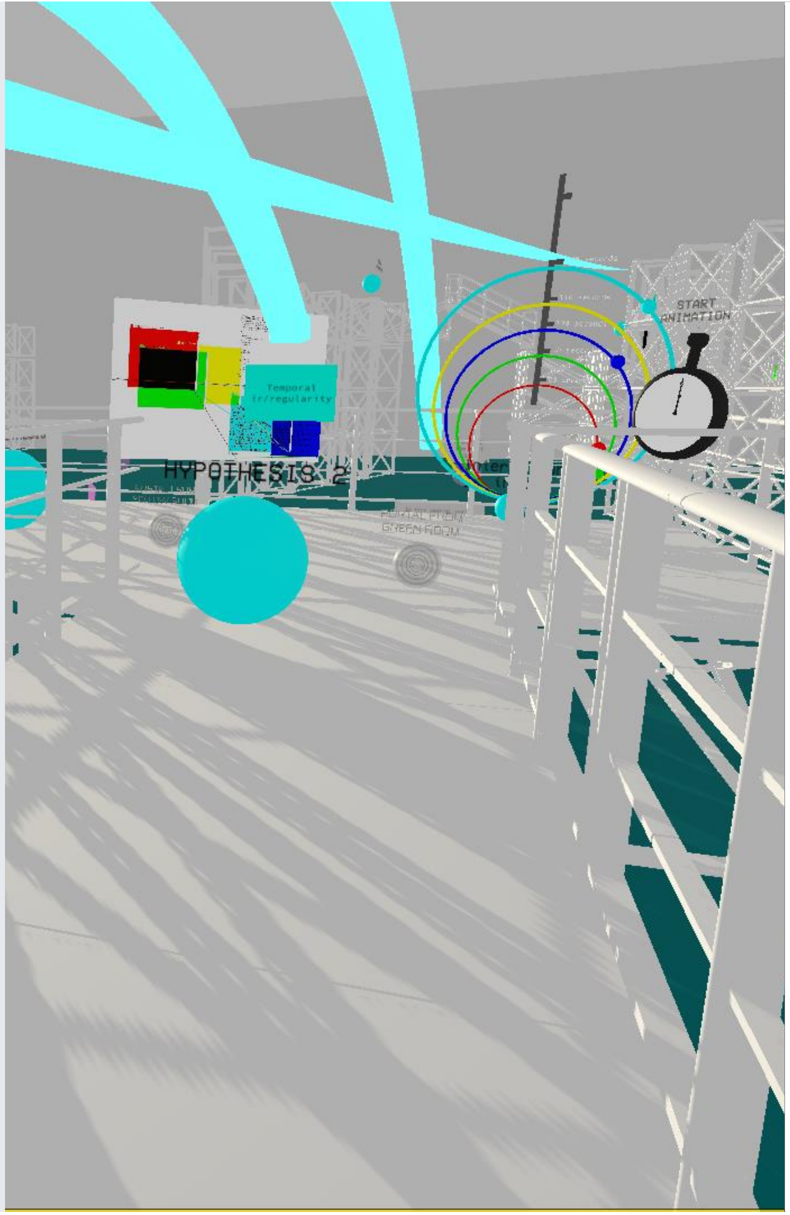
With a standard full HD screen, k being the value of the horizontal position, and 14j the random integer between 0 and 400, then k=1920-14j, 1520<k<1920. In this case, the middle-bar-URL-ball for the new URL-call will appear only between the 1520 pixels to 1920 pixels area of the screen as marked in magenta for Fig. 8a.

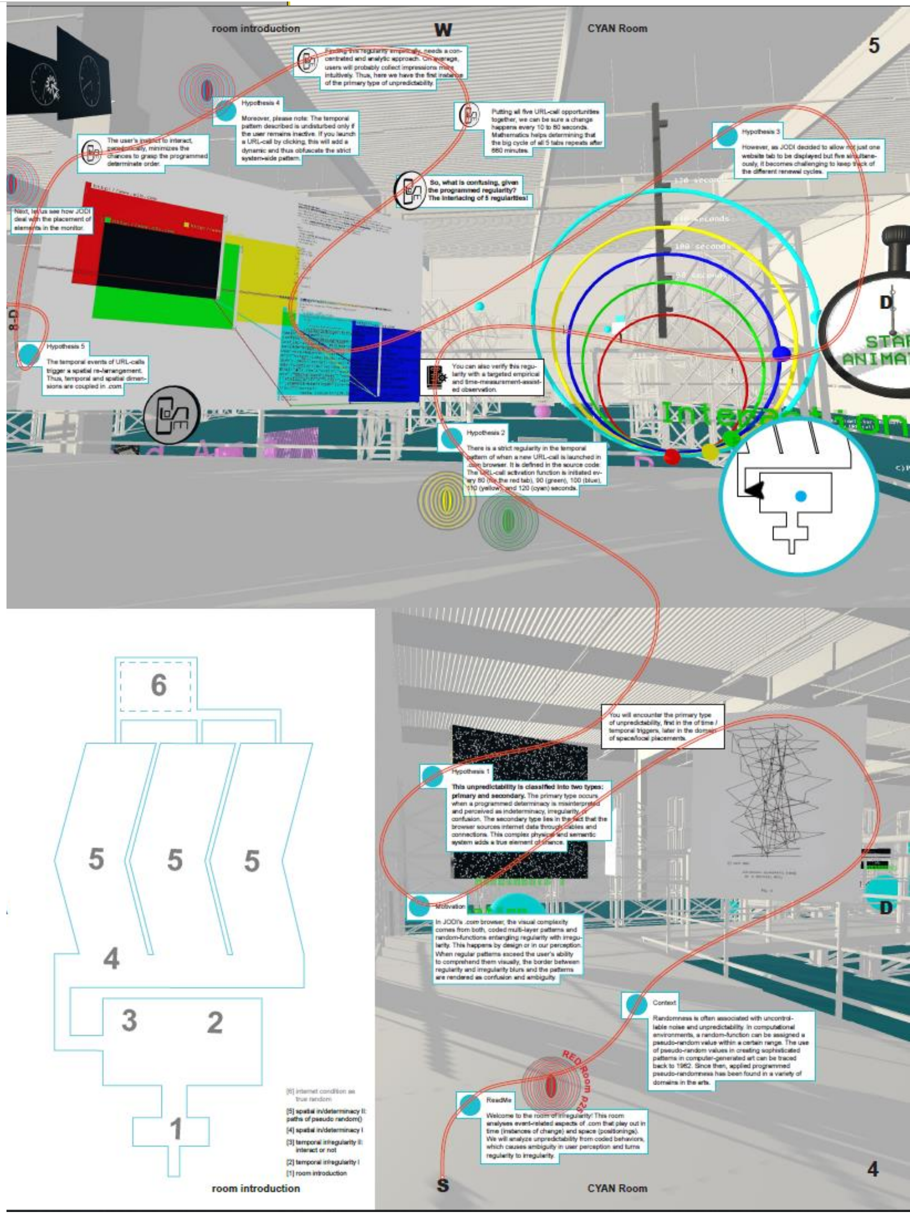
```
set the loch of sprite 86 to ((the loch of sprite 96) - 581)
set the locv of sprite 86 to ((the locv of sprite 96) - 10)
set the loch of sprite 87 to ((the loch of sprite 97) - 581)
set the locv of sprite 87 to ((the locv of sprite 97) - 10)
set the loch of sprite 88 to ((the loch of sprite 98) - 581)
set the locv of sprite 88 to ((the locv of sprite 98) - 10)
set the loch of sprite 89 to ((the loch of sprite 99) - 581)
set the locv of sprite 89 to ((the locv of sprite 99) - 10)
set the loch of sprite 90 to ((the loch of sprite 100) - 581)
set the locv of sprite 90 to ((the locv of sprite 100) - 10)
```

Fig. 8: A reworked screenshot after all the middle-bar-URL-balls have been reshuffled from their initial positions, the magenta rectangle marking the possible area for middle-bar-URL-ball to settle set by the random-function in a full HD size screen (8a). Corresponding code from the Main Movie Scripts (8b).

The moment the middle-bar-URL-ball's [13] horizontal position is set, the position of the formatted-text ([3], "tt1") is determined as well. While the formatted-text always starts at the top of the monitor, its left border is 576 pixels to the left of the corresponding coloured middle-bar-URL-ball. Due to the middle-bar-URL-balls' restricted settling area (Fig. 8a), the formatted-texts are also displayed on the right side of the screen. This provides an

43





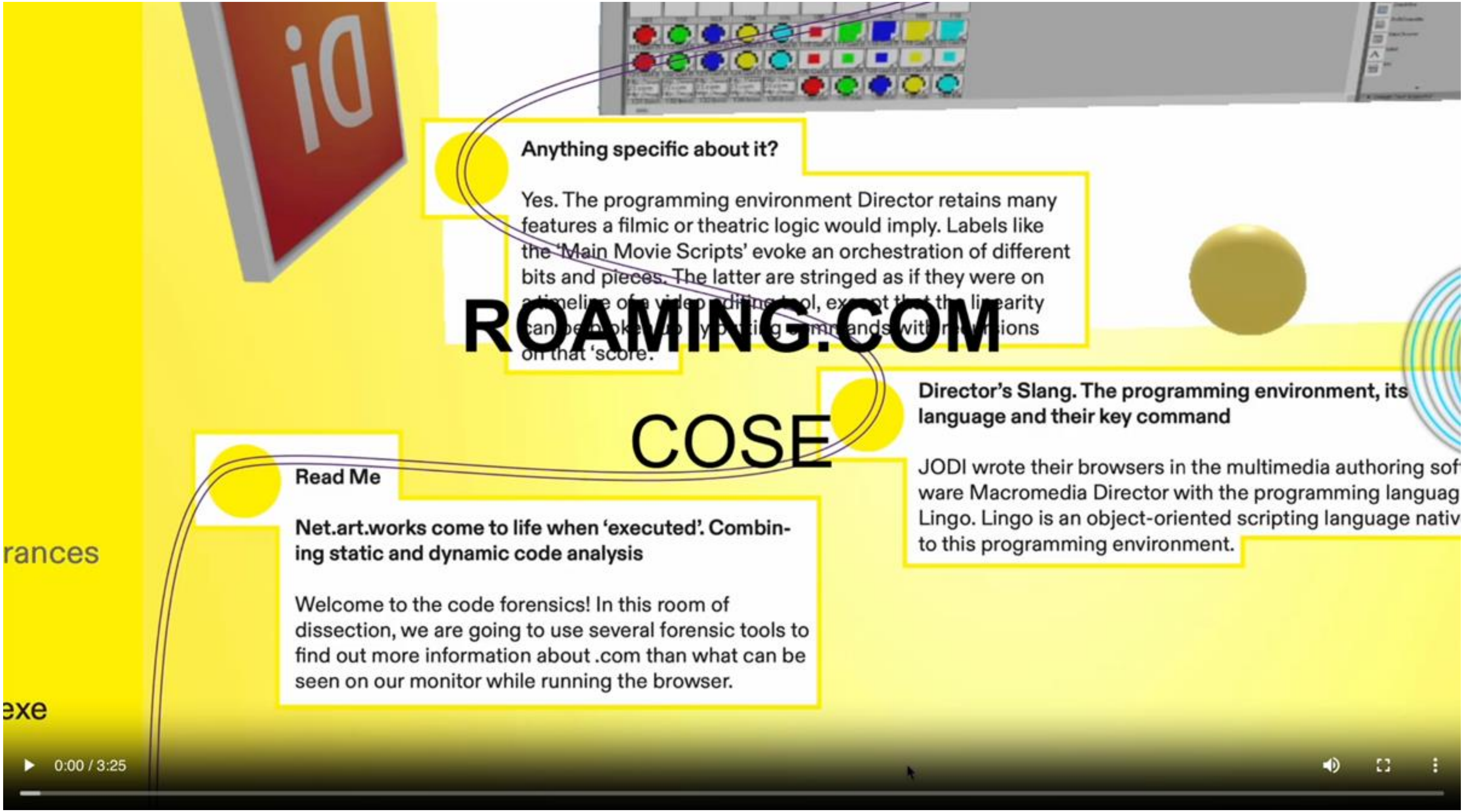
In Depth: Roaming Around the Conceptual Space of JODI's .com Browser
book

ROAMING.COM
game

ROAMING.COM printed
game guide

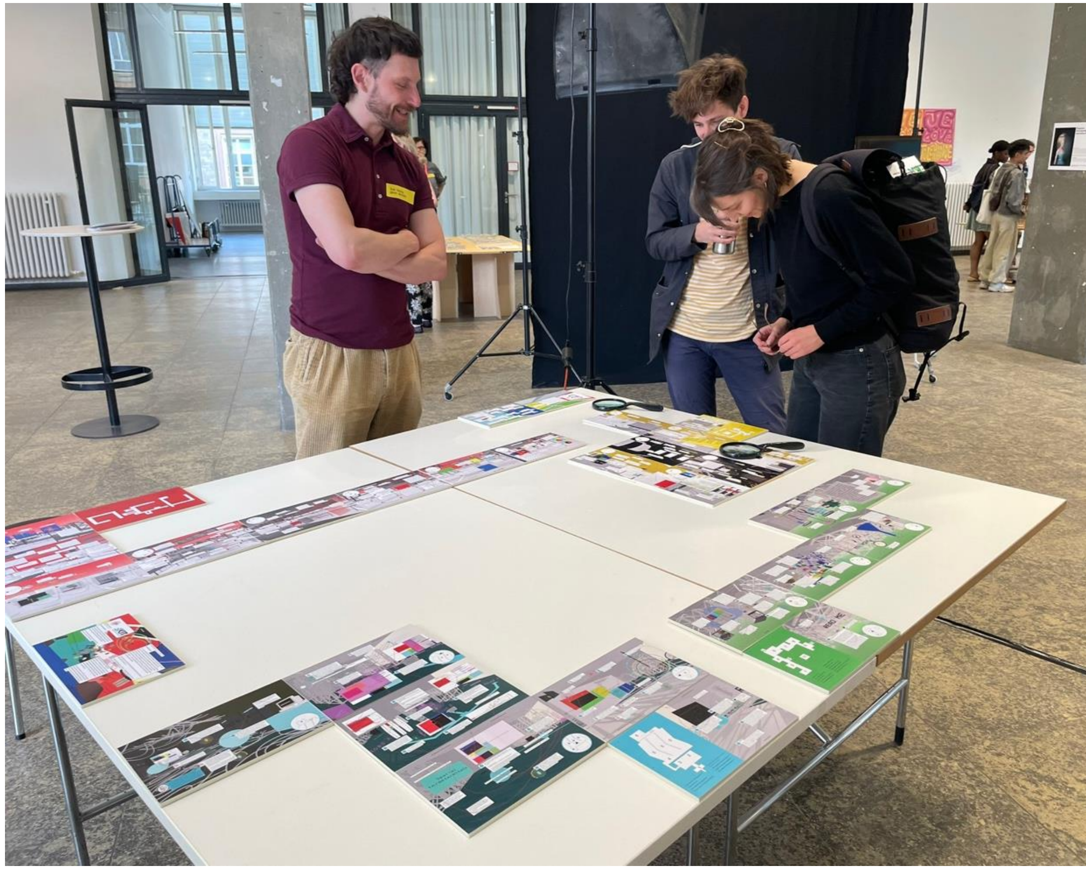
… and the resulting output: COSE's triple publication (2025/2026) + exhibition

ROAMING.COM, 2025.
Unity Game,
Analytic deep-dive



COSE: ROAMING.COM (2025) Demo narrated by Inge Hinterwaldner

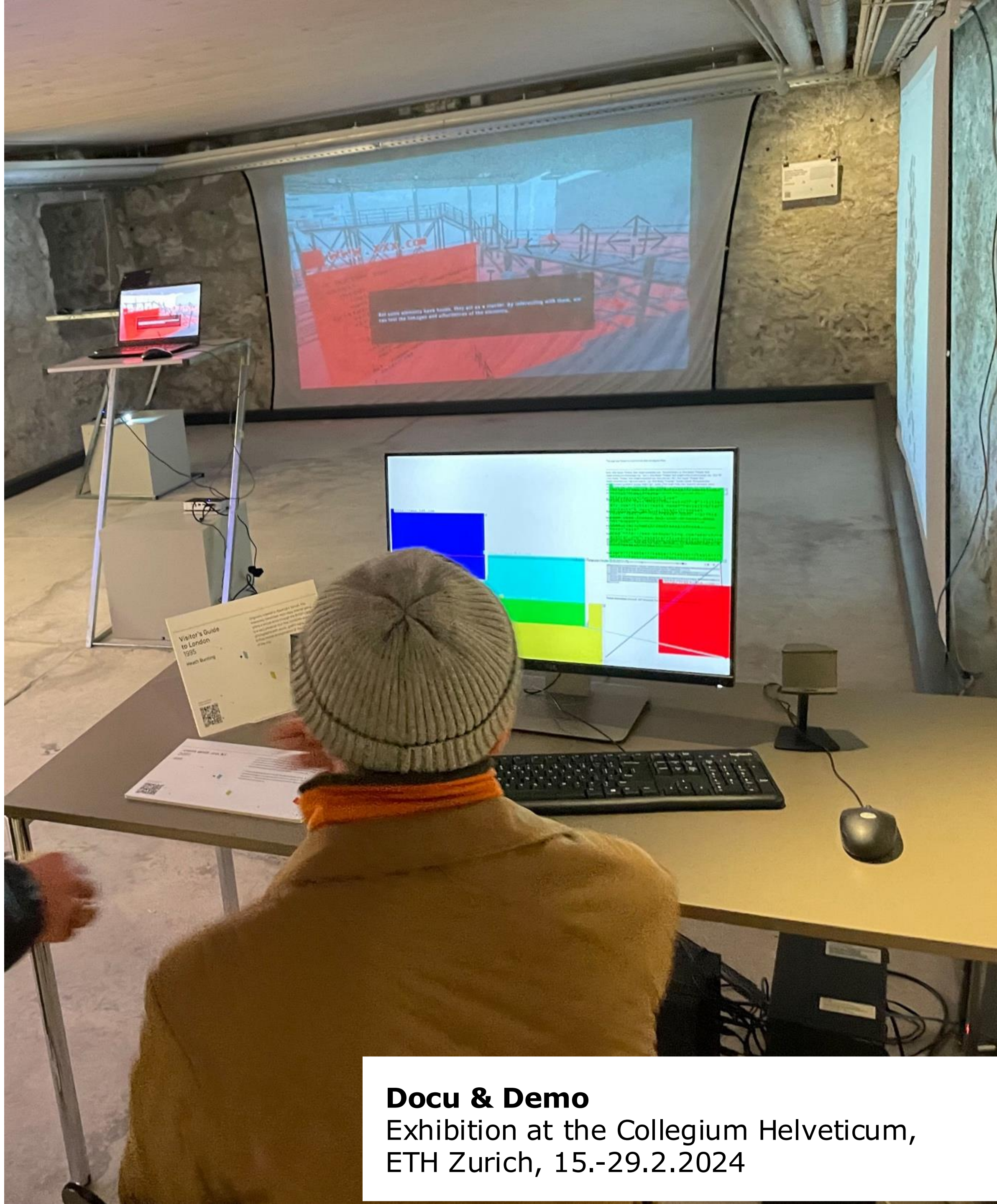
Third Mission



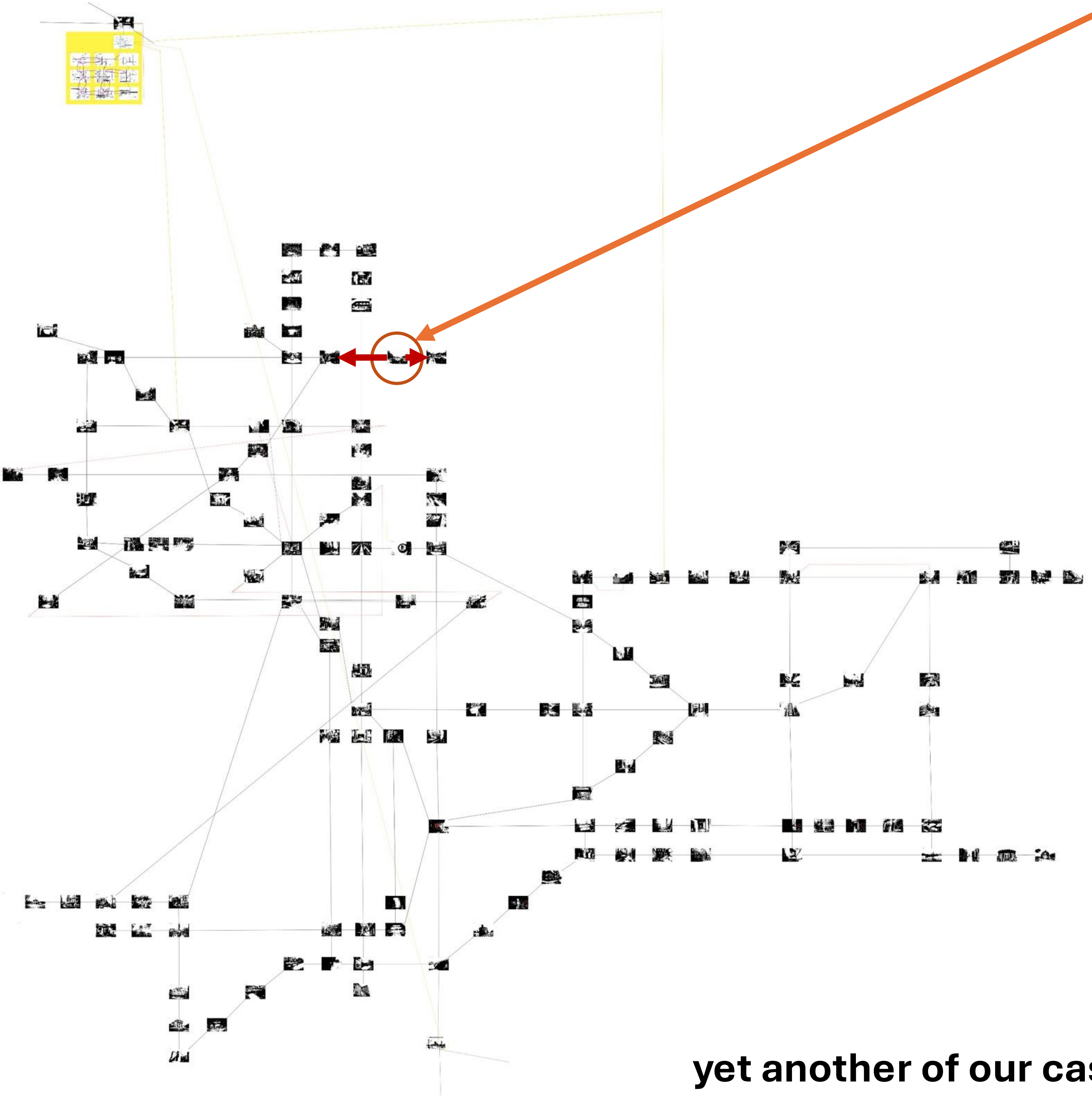
COSE: ROAMING.COM game guide version, 2025.

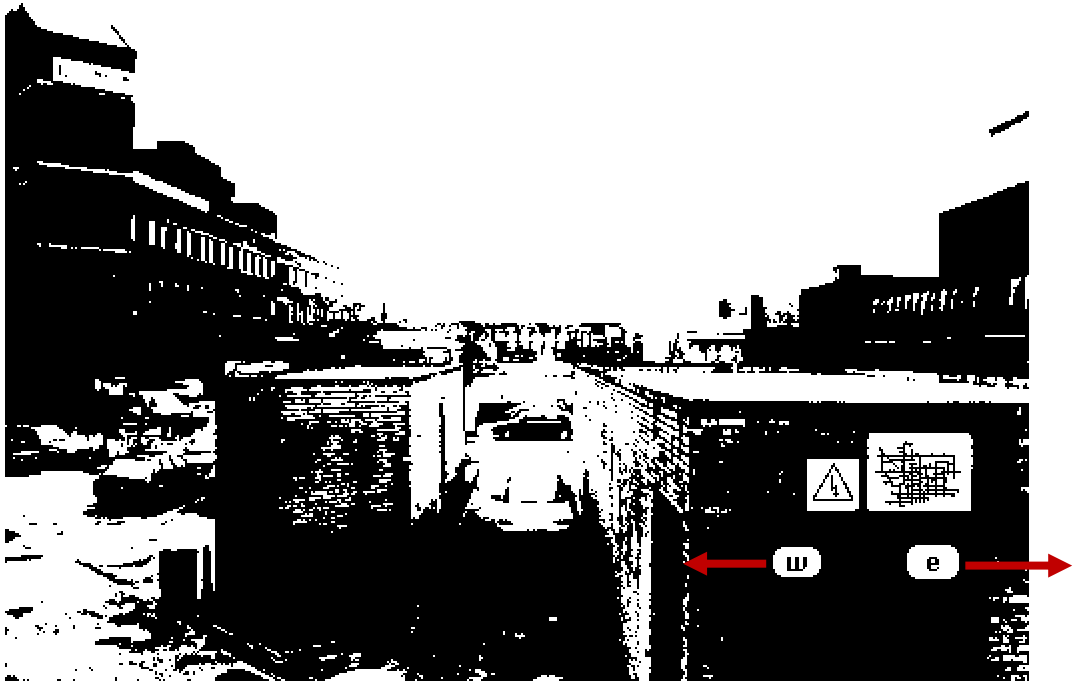
Storytelling: 4 argumentations, cardboards on table, magnifying glasses.

academic + public context



Docu & Demo
Exhibition at the Collegium Helveticum,
ETH Zurich, 15.-29.2.2024





Heath Bunting: *Visitor's Guide to London*, 1995.
Interface (= detail), is a massively hyperlinked website labyrinth

COSE: Second Order *Visitor's Guide to London*, 2024.
Map layer 'street view' (= detail photos, manually arranged along cardinal directions as indicated)

yet another of our case studies ...

Analyses of a highly networked artistic website

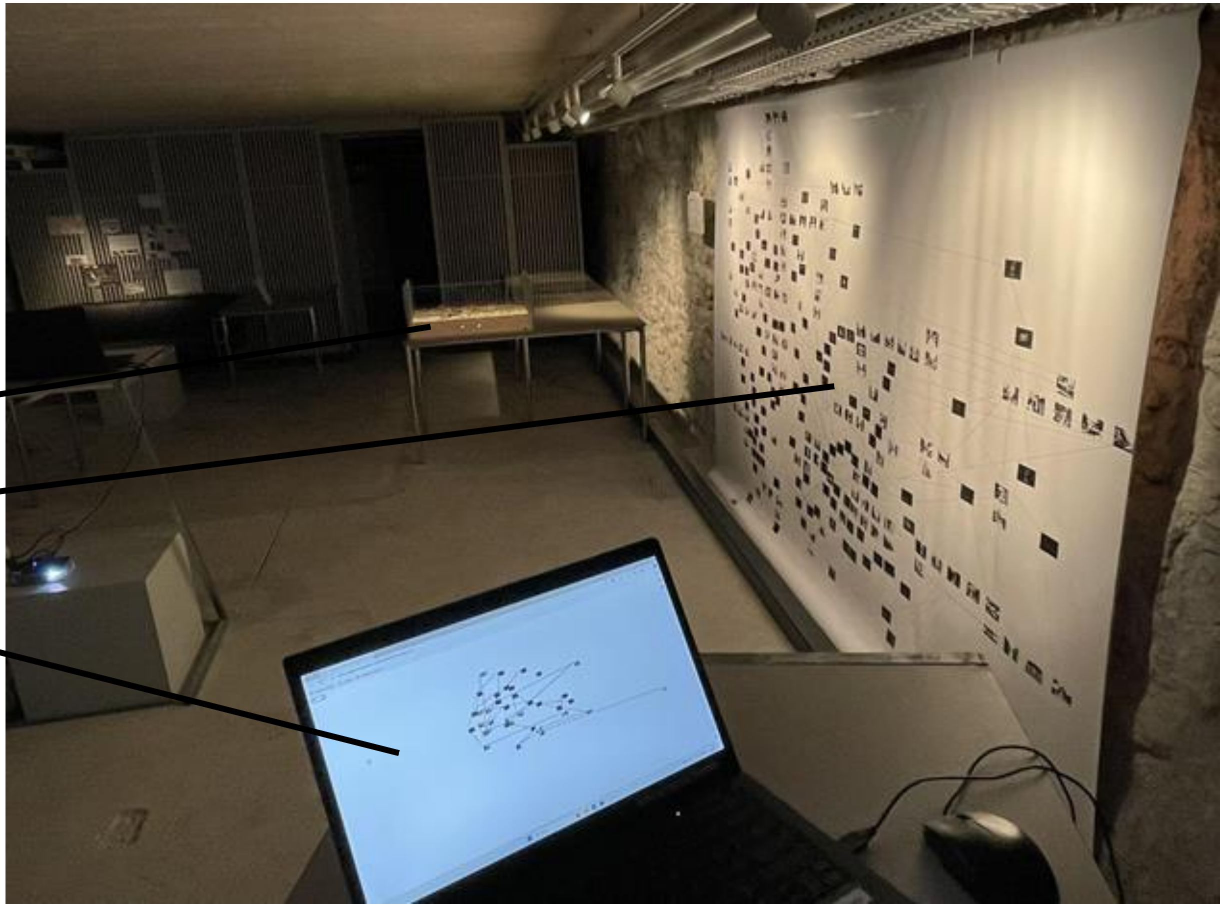
case studies

COSE: *Second Order* Visitor's Guide to London (2024)
(analysis of Heath Bunting: *Visitor's Guide to London*, 1995)

... and how it can be mapped and interpreted in various ways ...

(curated research-based) exhibitions

Docu & Demo: Archiving Programmed Media Art, Collegium Helveticum, ETH Zurich, Feb 15-29, 2024



case studies

Model set: physical 9-layer model, 3-layer poster, interactive mapping

Maps (poster, layer, dynamic) + source/reference
(COSE: *Second Order VGTL*, 2024)

Book + source/reference
(Browser Art: *Navigation,* 2022)

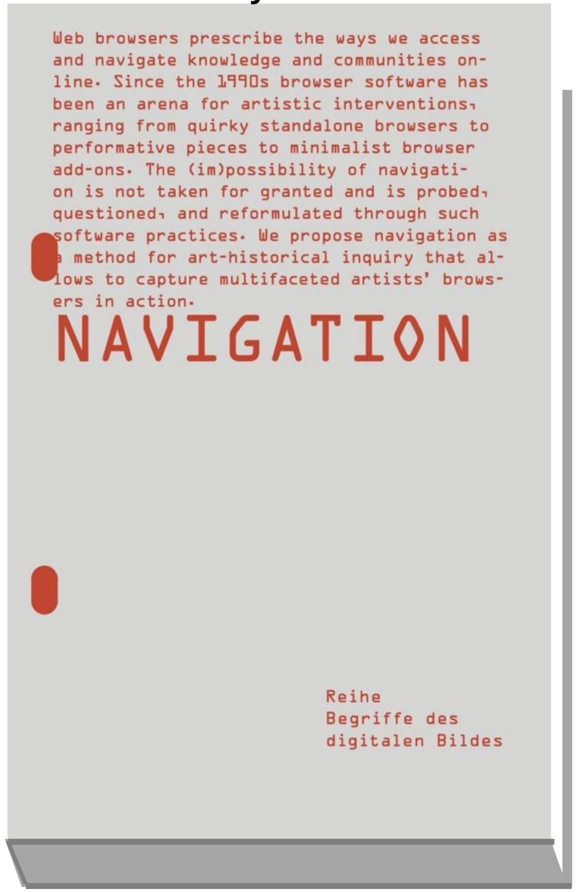
Argumentation (game, game guide, book) + source/reference
(COSE: *ROAMING.COM*, 2025)

3:3

(exhibitable) epistemic model complexes

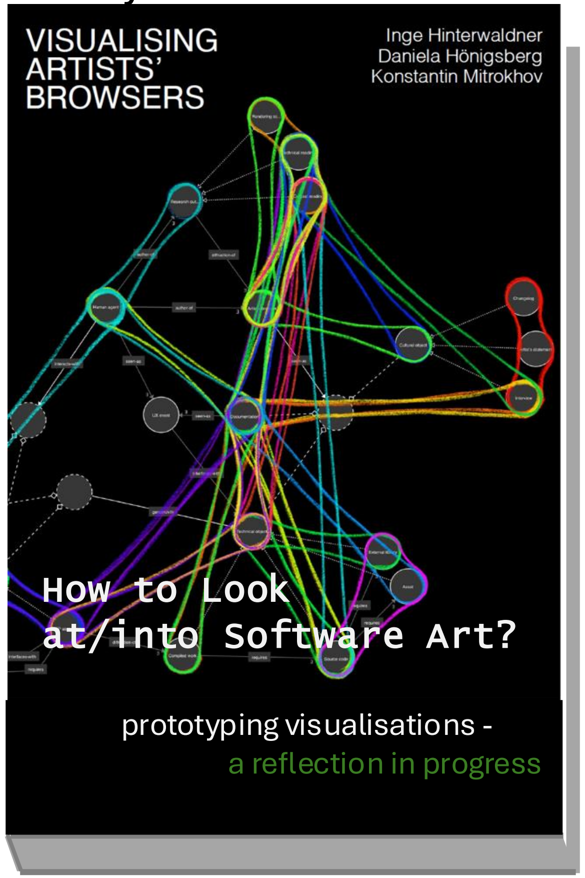
Book publications (ecosystem of publications around browser art)

method/documentation
case study



eds. Inge Hinterwaldner, Daniela Hönigsberg, Konstantin Mitrokhov,
130 p.
LMU Press: Munich 2022.

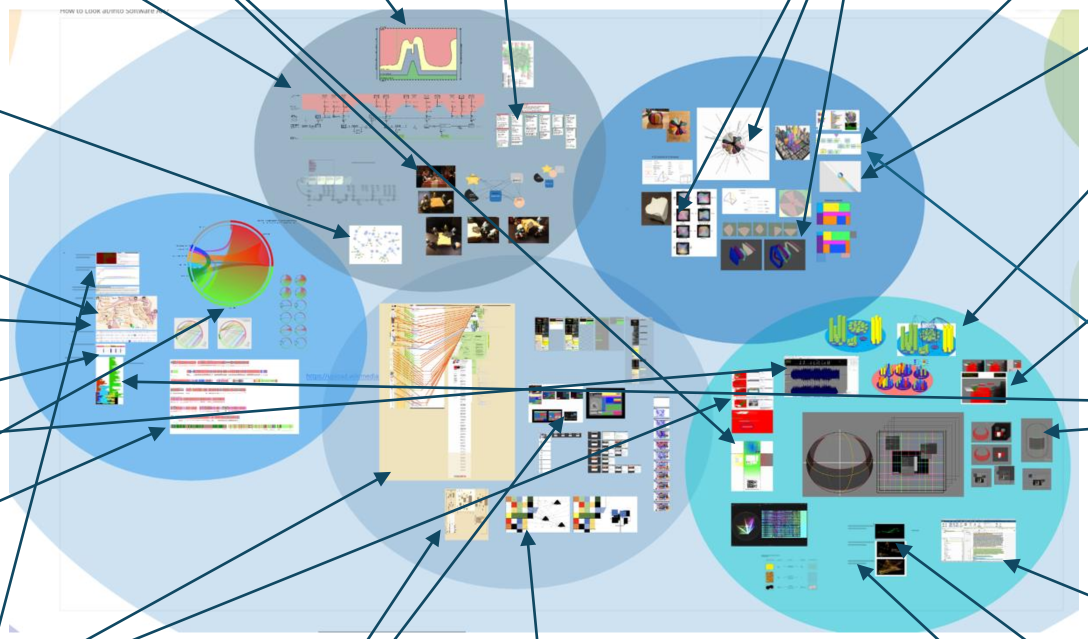
method/visualisation
survey

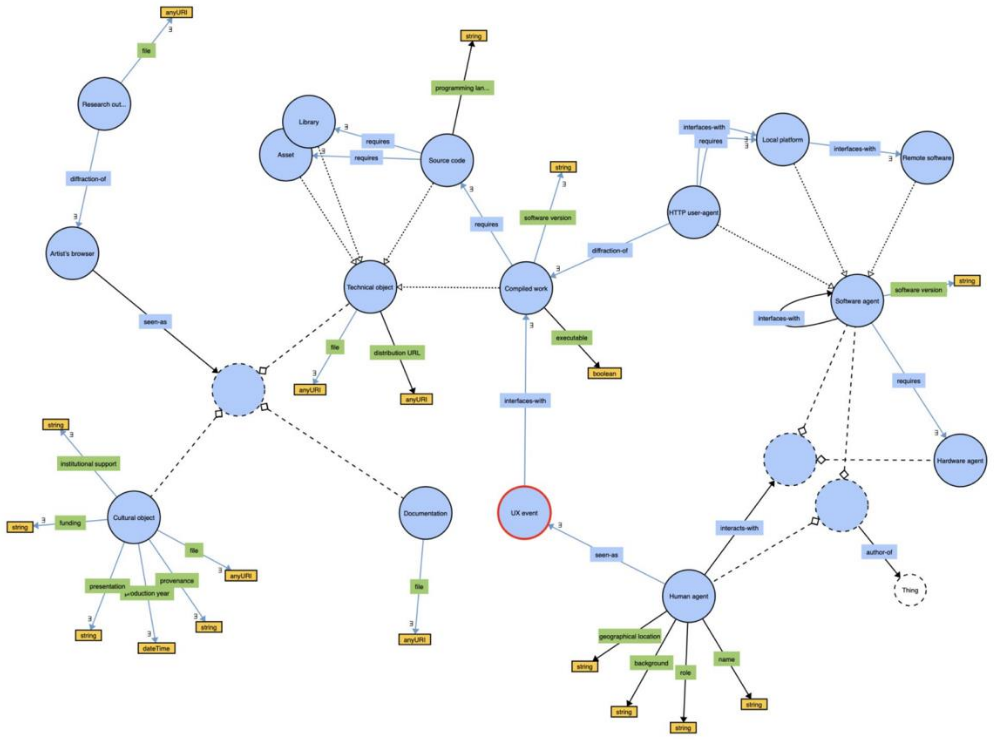


eds. Inge Hinterwaldner,
2021-2022 finished; conceptually a work in progress, 172 p.
Intercom: Zurich

20+ distinct *types* of visualisations (2020-2022)

… in progress …





SOFTWARE ONTOLOGY GRAPH

date: June/November 2021

status: finished (second iteration)

software used: WebProtégé, owl2vowl, WebVOWL (local instance)

DESCRIPTION

Context: this visualisation is part of an effort to model a software ontology of artist's browser.

Function: heuristic.

Intention: modelling a bottom-up ontology of artist's browser as a situated software class that includes multiple research perspectives.

Question: is the ontology accurate and full?

Source of data: embodied knowledge of the modelled subject.

Sequence: 1) model the ontology in WebProtégé through an iterative process with the aim of accounting for most relations, subclasses, and research perspectives; 2) export the file from WebProtégé in the format supported by WebVOWL; 3) load the file into WebVOWL running locally in a supported browser; 4) tweak the graph display parameters for better legibility with the help of "pin" tool; 5) revise the graph and make notes for the next iteration of the software ontology; 6) repeat steps 1-5 until the ontology is satisfying.

Sources & references: ontology modelling, knowledge graphs.

BRIEF REFLECTION

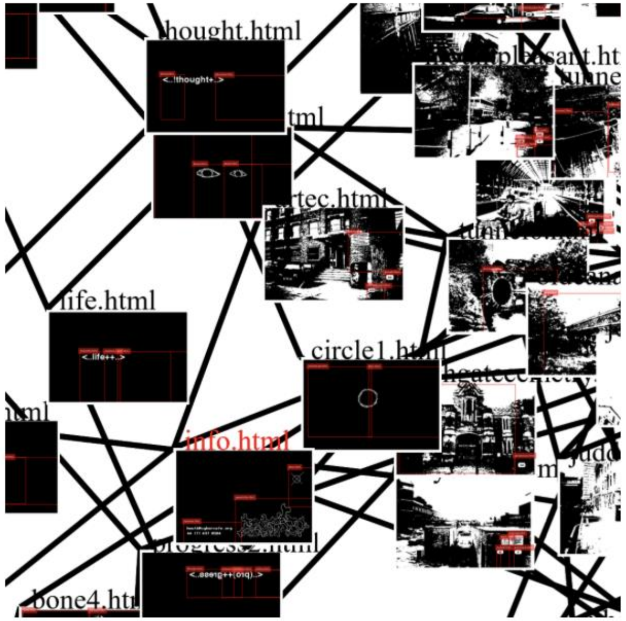
This graph is extremely useful during the process of modelling an ontology. WebProtégé lacks built-in visualisation tools and doesn't provide a visual overview of modelled ontologies. It is difficult to have an overview of an ontology without looking at its graph that shows all the relations and annotations. When the ontology is finished, its graph could serve as a baseline vector illustration for further design work.

The main issue with ontology modelling is that its graph is as abstract as all other graphs.

Analytical tools

NAD | Net Art Detector Tool v0.01, 2024

The NAD Tool is a browser0based application developed by Daniela Hönigsberg to support the investigation and analysis of web-based artistic pratices and to assist researchers, curators, and students in identifying webpages that diverge from conventional design practices. Read more …



Heat Map Web Crawler with Visual Site Mapping, 2024

The Heatmap Web Crawler with Visual Site Mapping tool is a desktop application that crawls websites and generates visual site maps with screenshots. This software tool is built with Electron, Express, and Mermaid.js. by Emma Dickson. Read more …

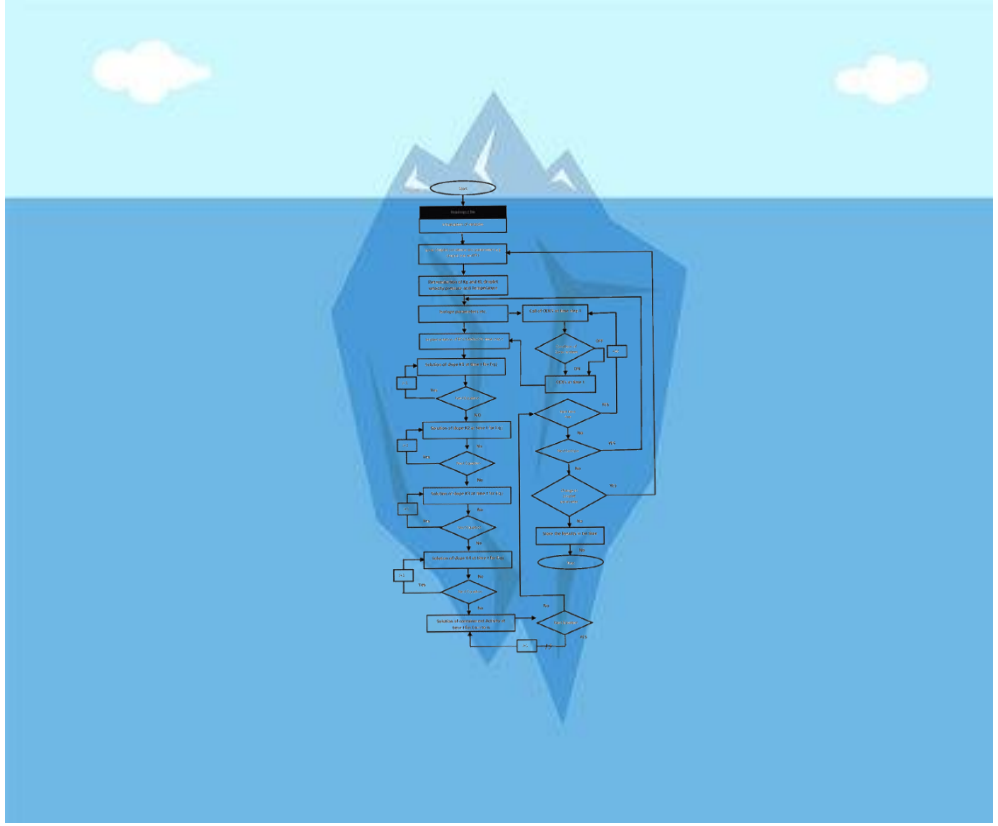
Source: www.codedsecrets.eu

Functional portraits

(desire, not done yet on a larger scale)

Expand the levels of comparison between artworks based on an ontology

Future work: creating artworks-specific portraits of how they function



monitor output
(phenomenological level)

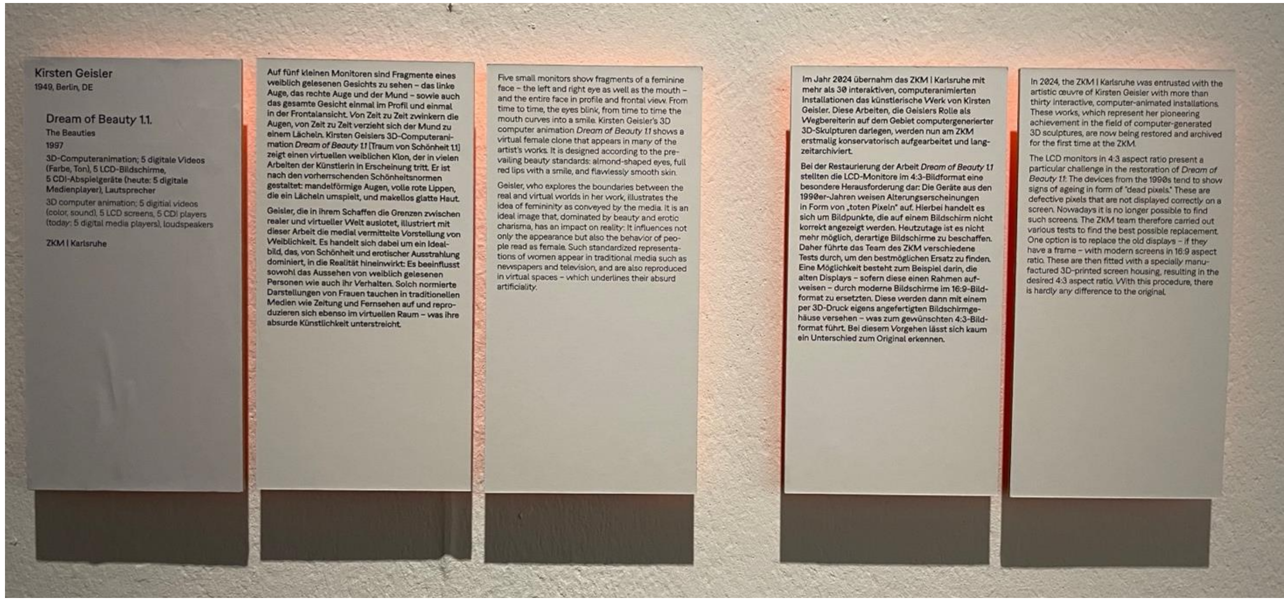
program execution
background processes
functionality

Vision: exhibitions as ready-made experimental setups 4 systematically logging the changes

- Build a community of curators / artists / technologists / conservationists interested in exhibiting online art
- Conceive exhibitions as occasions to rethink living-on-formats (online, offline, hybrid)
- Curators / artists conceptualise the next iteration
- Conceive exhibitions as occasions to log software changes
- Technologists / conservators update the works to make them ready for the show
- Technologists / conservators repair the works while on show
- They record what was done why and submit entries to a crowd source interface
- Community of curators / technicians / conservators / artists publish jointly in a quarterly (edited sum of entries)

Benefits:

- artwork-relevant record of online events that else go unnoticed (humanities‘ interest, history)
- solutions published beneficial for all other works suffering that were not on show at that time (artists /museums interest)
- destigmatise the fact that artworks need repair (conservators‘ interest, whose work becomes more visible)



ZKM exhibition “The Story that Never Ends”, since 2025-2028.

Thank you for your attention!

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